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SELF-SERVICE PRINTING STATION IN SKSU – ISULAN LIBRARY WITH CASHLESS PAYMENT INTEGRATION VIA GCASH

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TRANSMITTAL

The thesis attached hereto entitled "**SELF-SERVICE PRINTING STATION IN SKSU – ISULAN LIBRARY WITH CASHLESS PAYMENT INTERGRATION VIA GCASH"** prepared and submitted by **NATHANIEL C. LAGNAODA** and **GEORGE ANDRIE D. GATAN,** in partial fulfillment of the requirements for the degree *Bachelor of Science in Computer Engineering* is hereby endorsed for approval.

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BIOGRAPHICAL DATA

The researcher was born on September 22, 2003, in General Santos City he is youngest child of Mr. George Y. Gatan and Mrs. Marry Ann D. Gatan. He is fondly called “Adi,” “Gats,” and “Drei” by his family and friends.

He spent his childhood in his birthplace and completed his elementary education at Montessori Casa Precious Jewels Inc, Sto Niño, South Cotabato, in 2016. He persued his junior high school at Montessori Casa Precious Jewels inc and graduated in 2019-2020.For his senior high school, he studied at Sto Niño National High School under the Science, Technology, Engineering, Mathematics (Stem), which he completed in 2022. At present, he is pursuing a Bachelor of Science in Computer Engineering at Sultan Kudarat State University, Isulan Campus, Where he continues to enhance his technical and analytical skills

Looking ahead, the researcher aims to continue improving himself as he faces new opportunities and challenges with determination, hard work, faith in God, he hopes to achieve in his chosen career in the field of technology, possibly focusing in software development, systems design, or automation.

BIOGRAPHICAL DATA

The researcher, Nathaniel C. Lagnaoda, was born on December 12, 2003, at Quijano Clinic and Hospital in Tacurong City. He is the son of Mr. Bonifacio T. Lagnaoda Jr. and Mrs. Belinda C. Lagnaoda and is fondly called “Nikol” by his family and friends. He grew up with a curious mind and a strong passion for technology, showing early interest in computers and design. His upbringing instilled in him the values of hard work, perseverance, and humility, which continue to guide him in both his academic and personal life.

He began his educational journey at Masiag Central Elementary School, where he completed his elementary education. He then pursued his junior and senior high school studies at Notre Dame of Masiag, Inc., where he further honed his academic foundation and leadership skills. Currently, he is a fourth-year student taking up the Bachelor of Science in Computer Engineering at Sultan Kudarat State University, Isulan Campus. Throughout his college years, he has developed skills in programming, AutoCAD, circuit simulation, and basic hardware design using Arduino and ESP32 microcontrollers. He is also an active student leader, serving as the Business Manager of the Engineering Student Organization, Class Representative of ICpEP.SE, and President of the Book Club.

Looking ahead, Nathaniel aspires to become a skilled and innovative computer engineer who can contribute to the advancement of technology and make a meaningful impact in his community. He plans to specialize in fields such as network engineering, automation, or embedded systems development. With dedication, continuous learning, and faith in God, he envisions building a successful career in the tech industry and possibly establishing his own technology-based enterprise to help empower others through innovation and education.

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Researchers

NATHANIEL C. LAGNAODA

GEORGE ANDREI D. GATAN

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ABSTRACT

NATHANIEL C. LAGNAODA and **GEORGE ANDRIE D. GATAN**, **Month And Year Published**. "**SELF-SERVICE PRINTING STATION IN SKSU-ISULAN LIBRARY WITH CASHLESS PAYMENT INTEGRATION VIA GCASH**". A thesis of Bachelor of Science in Computer Engineering, Sultan Kudarat State University, Isulan Campus, Isulan, Sultan Kudarat.

ADVISER: JADE A. BUENAVIDES

In this study, the researchers developed and implemented a Self-Service Printing Station integrated with cashless payment via GCash, designed to provide students with a faster, more convenient, and efficient printing experience within the SKSU–Isulan Library. The station automates the document printing process by allowing users to upload their files, convert them to PDF, preview the document, and pay electronically through GCash before printing. The system utilizes a Raspberry Pi microcontroller, touchscreen monitor, and printer, all integrated through a custom software interface for smooth operation.

The project aims to address the challenges of long queues, manual payments, and the lack of accessible printing facilities within the library. Test results and observations show that the system successfully streamlines the printing process, minimizes human intervention, and promotes a secure, contactless transaction environment. This innovation enhances operational efficiency in the university library and aligns with modern digital payment trends, providing a sustainable and user-friendly solution for students and staff alike.

Chapter I

INTRODUCTION

Background of the Study

Educational institutions today face an increasing demand for printing and document handling services from students, faculty, and staff. Traditional manual print stations often result in delays, long queues, and inefficient workflows, particularly during peak academic periods. In the context of the Sultan Kudarat State University Isulan Campus library, the absence of an automated, cashless, self-service printing station has led to bottlenecks, extended waiting times, and greater dependence on library staff to manage printing and payment transactions. These challenges highlight the need for a more efficient and user-friendly solution to meet the academic community's printing requirements.

The effectiveness of self-service station systems in improving process efficiency and user satisfaction has been demonstrated in various international studies. Kim, Yang, and Lee (2023) reported that the adoption of self-service stations in the Republic of Korea enabled organizations to improve operational resilience during the COVID nineteen pandemic by reducing physical contact, supporting cashless transactions, and accelerating service delivery. Similarly, Chung and Park (2021) evaluated the usability of self-service stations among older adults and found that design features such as seating and spatial partitions significantly reduced user workload and enhanced task performance. These findings support the view that introducing self-service stations can improve service delivery, enhance user experience, and streamline operations.

In the Philippine context, mobile based and station-based innovations are increasingly gaining attention. A local study on customer perceived satisfaction with self-service stations in fast food establishments found that ease of use, navigability, and time saving features had significant positive effects on user satisfaction (Jherald et al., 2025). In addition, a report by Office (2025) on the first self-service printing station in the Philippines highlighted how print vending stations provide convenience, privacy, and accessible payment options while generating additional revenue and reducing labor costs for operators. These local insights demonstrate the applicability and potential of self-service printing stations within the Philippine setting.

Within the Mindanao region, there is growing institutional interest in self-service systems. For instance, Bukidnon State University (2023) introduced a self-check station for library transactions to improve operational efficiency and user convenience. At the University of Southern Mindanao, Tutor (2025) reported that a print on demand system was benchmarked by other institutions in Davao del Sur as a cost effective and flexible digital printing service for academic use. While these developments indicate readiness for print service innovations in Mindanao, they still lack fully automated, cashless, self-service printing stations integrated with mobile wallet payment systems.

The effectiveness of a self service printing station integrated with a cashless payment system lies in its ability to reduce manual transaction handling, shorten waiting times, enhance user satisfaction through convenience, and provide administrators with usage and sales reports for monitoring and decision making. Allowing users to independently upload documents, pay through mobile wallets such as GCash, and receive printed output with minimal staff assistance aligns with modern academic service delivery and digital transformation initiatives.

Despite continuous efforts toward technological advancement, Sultan Kudarat State University has yet to adopt a self service printing station integrated with a secure cashless payment system within its library. This gap underscores the need for a localized solution tailored to the operational context of the SKSU Isulan Library.

Therefore, this study aims to design, develop, implement, and evaluate a self service printing station in the SKSU Isulan Library that integrates a cashless payment system through GCash.

Objectives of the Study

General Objectives

This study generally aims to develop a self-service printing station in SKSU-Isulan library that provides students and users with a convenient, automated, and cashless payment service through GCash.

Specific Objectives

1. Design a web-based system for document handling and printing.

2. Allow platform for seamless and cashless transaction through Gcash.
3. Generate data analytics for sales, transaction logs, and usage.
4. Evaluate the system in terms of:
    4.1. Functionality
    4.2 Reliability
    4.3 User Convenience

Significance of the Study

This study is designed to develop and implement a Self-Service Printing Station in the Sultan Kudarat State University Isulan Campus Library, integrated with a cashless payment system through GCash. The primary goal is to provide students, faculty, and staff with a convenient, automated, and secure method of printing documents without the need for manual transactions or cash handling. The system contributes to promoting digital transformation and technological innovation in campus services. Specifically, this study is significant to the following:

Students. This system offers convenience and efficiency to students by allowing them to print documents anytime without relying on library staff. Through cashless payment via GCash, students can transact quickly and securely, saving both time and effort. The station helps address issues of long queues and manual payment processes, enhancing their overall learning and library experience.

Library Staff and Administrators. The station automates printing and payment operations, minimizing manual supervision and reducing workload. It also provides automated transaction logs, weekly and monthly reports, and usage summaries that can be used to monitor revenue, system utilization, and performance. This enables better management and transparency in library services.

University Management. The project demonstrates the institution's commitment to digital innovation and modernization. By adopting an automated and cashless system, the university promotes environmental sustainability through paperless payment and resource efficiency. It also enhances SKSU's reputation as a forward-thinking university that embraces financial technology and automation for public service improvement.

Researchers. This study is significant to the researchers as it enhances their technical and analytical skills in the field of computer engineering. Through the design and implementation process, the researchers gain hands-on experience in system integration, hardware and software interfacing, and payment gateway development. It also strengthens their knowledge of user-centered design, cybersecurity, and system evaluation.

Future Researchers. This study may serve as a useful reference for future researchers who wish to explore similar innovations involving self-service technology, automation, and financial technology integration. The framework, methodologies, and findings presented in this research can guide them in developing more advanced and efficient systems, particularly those that aim to improve public access and service automation in academic or institutional settings.

Scope and Limitation

This study focuses on the development and evaluation of a web-based system that automates the complete printing workflow within a library setting. The system is designed to accept document uploads only in PDF, DOCX, and standard image file formats, with a maximum file size of 25 MB per document. A fixed pricing scheme is embedded in the system: 2 pesos per page for Letter and A4 sized black and white prints, 5 pesos per page for Legal sized black and white prints, and a flat rate of 4 pesos per page for all colored prints regardless of paper size. Cashless payment processing is implemented exclusively through the Xendit API.

The implementation and evaluation of the system are conducted at the SKSU Isulan Library, with the primary users being the library patrons. In line with the research design, respondents are selected using purposive sampling, targeting users who directly interact with the system. The system is evaluated based on functionality, reliability, and user convenience using User Acceptance Testing, system generated logs, and structured survey instruments.

To maintain feasibility and alignment with the study objectives, several limitations are imposed. The system is limited to a responsive web interface and does not include a dedicated mobile application; consequently, access to the system is restricted to users with smartphones capable of scanning QR codes to initiate transactions. File support is limited to the specified formats and size constraints, excluding larger or unsupported files. The pricing model is fixed and does not allow dynamic pricing or promotional adjustments. The study utilizes existing or commercially available printing hardware and does not assess custom or specialized devices. Furthermore, the implementation is geographically confined to the SKSU Isulan Campus Library and is not extended to other SKSU campuses. Methodologically, the findings are limited to data collected from purposively selected users and may not be generalized beyond the study context.

Operational Definition of Terms

The following terms are operationally defined as used in this study:

Administrator - Refers to the person responsible for managing, maintaining, and monitoring the printing station system, including generating reports of sales.

Cashless Payment - Refers to the incorporation of electronic or digital payment systems into the printing station, allowing users to complete transactions without using physical cash, specifically through GCash or similar mobile wallet services.

Database - Refers to the structured data storage system that records transaction logs, payment histories, and usage reports for administrative monitoring and analysis.

Data Analytics - Refers to the process of collecting, analyzing, and interpreting printing station's data such as transaction logs, daily usage, and sales reports.

Self-Service - Refers to a system or process that allows users to perform tasks or access services independently without the need for assistance from staff or personnel.

Functionality - Refers to the ability of the system to perform its intended operations effectively, such as printing, payment processing, and data recording without errors or failures.

GCash - Refers to a mobile wallet application in the Philippines that allows users to pay bills, send money, and make purchases online or in-store using their smartphones.

Printing Station - Refers to an automated system or station designed to provide on demand printing services. It allows users to upload, preview, and print documents independently without requiring staff assistance.

Raspberry Pi 4 - Refers to the small, single-board computer that serves as the main processing unit of the self-service station, managing both hardware and software operations.

Reliability - Refers to the consistency and dependability of the system in performing its tasks, ensuring that printing and payment processes function accurately and continuously without frequent breakdowns or errors.

SKSU-Isulan Library - Refers to the designated location within the Sultan Kudarat State University – Isulan Campus where the printing station is installed and utilized by students and staff.

System Interface - Refers to the graphical layout and interactive components that enable users to communicate with the printing station system, including display menus, buttons, and prompts for uploading files and confirming payments

User Convenience - Refers to the level of ease, accessibility, and satisfaction experienced by the users while interacting with the system, particularly in printing documents and making payments efficiently.

Xendit - Refers to the payment gateway platform that is integrated into the printing station system that facilitates secure and real-time transaction processing between GCash and the printing system.

CHAPTER II

REVIEW OF RELATED LITERATURE

The review of related literature and studies will be covered in this chapter. By giving insight into the characteristics of people that are fundamentally necessary to the study's title, "Self Service Printing Station in SKSU-Isulan Library with Cashless Integration via Gcash”.

Raspberry Pi

The Raspberry Pi was selected as the main controller of the self-service printing station due to its sufficient processing capability, Linux-based operating system support, and compatibility with peripheral devices and network-based applications. Its General Purpose Input/Output (GPIO) pins allow seamless integration with printers, touchscreens, and other hardware, making it suitable for automation and embedded system applications. Studies have shown that the Raspberry Pi is widely used in engineering and research environments requiring real-time data processing and system control. Thakur et al. (2023) highlighted its effective application in areas such as industrial automation and machine vision, demonstrating its capability to handle both data acquisition and on-board processing.

Furthermore, the Raspberry Pi’s support for programming languages such as Python and system-level libraries enables efficient development of user interfaces, printing services, and API-based payment verification. Rani and Singh

(2022) emphasized that compared to traditional microcontrollers, the Raspberry Pi offers advantages in applications requiring networking, image processing, and cloud connectivity, as it operates as a full Linux-based computer while retaining hardware-level control. These characteristics make the Raspberry Pi an appropriate platform for implementing an automated printing station that requires file processing, payment validation, and reliable system coordination.

Inkjet Printer

Inkjet printers are commonly used in academic and small scale service environments due to their ability to produce high quality text and image outputs at a relatively low cost. Their compatibility with computer based printing systems and support for standard document formats make them suitable for automated printing applications. Studies indicate that inkjet printers provide sufficient resolution for academic documents and images while maintaining compact hardware requirements, which is ideal for station based installations (Hanna, 2022).

From a technical perspective, inkjet printing is a digital printing method that directly deposits ink droplets onto various substrates to recreate digital images. Shah et al. (2021) classified inkjet printing technologies and highlighted their wide applicability across multiple fields, emphasizing piezoelectric inkjet printing for its reliability and ability to handle diverse ink formulations. The study explained that piezo driven inkjet printheads operate by applying controlled voltage waveforms to a piezoelectric membrane, ensuring stable droplet formation and consistent print quality. These characteristics contribute to improved printing accuracy, stability, and speed, which are critical for repeated and automated printing tasks.

Similarly, Yesuf et al. (2023) noted that modern inkjet printers operate using either continuous inkjet or drop on demand mechanisms, where ink formulation and droplet control influence print accuracy and output quality. Maintenance mechanisms are incorporated to prevent nozzle clogging and ensure consistent performance during continuous use. In automated printing systems, inkjet printers support digital workflow integration through operating systems and print management services such as CUPS.

The reliability, affordability, and ease of maintenance of inkjet printers make them an appropriate choice for a self service printing station intended for student use. Their stable operation, acceptable print clarity, and efficient document processing support consistent service delivery in institutional environments such as university libraries.

Touch Screen Display

Touch screen displays have become a core component of modern digital systems due to their ability to support direct and intuitive user interaction. Sirois (2024) emphasized that touch-based interfaces enhance usability and efficiency by allowing users to interact directly with on screen elements, eliminating the need for external input devices. This interaction model has been widely adopted in self-service stations and other automated systems because it reduces user effort and improves overall accessibility.

Beyond basic touch input, advancements in touchscreen technology have focused on enhancing user experience through tactile feedback. Haghighi Osgouei (2020) explained that surface haptics technologies, such as electrostatic friction displays, improve touchscreen usability by modulating friction between the user's finger and the screen surface. This creates tactile sensations that allow users to both see and feel digital content, resulting in improved interaction accuracy and user satisfaction. The lightweight and electrical nature of this technology makes it suitable for integration into modern touchscreen devices used in automated systems.

In self-service applications, touch screen displays contribute to improved user experience by enabling clear navigation, real time feedback, and streamlined task execution. In the context of a self-service printing station, the touchscreen serves as the primary interface that guides users through document selection, print preview, and payment confirmation. The integration of responsive and feedback-oriented touchscreen technology supports independent operation, minimizes reliance on staff assistance, and enhances overall system efficiency.

Quick Response (QR) Code

QR codes are widely used in digital systems to enable fast and contactless interaction between physical interfaces and online services. Hayes (December 2025) noted that QR based technologies support efficient data access and transaction initiation by allowing users to scan codes using standard mobile devices, making them particularly suitable for payment verification and information retrieval applications. Their ability to encode various data types enables seamless integration with modern digital platforms, including cashless payment systems.

In automated service environments, QR codes enhance usability by simplifying user actions and reducing manual input requirements. Their adoption in digital payment and self-service stations demonstrates their effectiveness in improving transaction speed and user engagement. However, the increasing use of QR codes in financial transactions has also raised concerns related to security and system integrity. Sanapala and Gondi (2023) emphasized that QR code-based payment processes interact with multiple digital components, including mobile devices, applications, and payment gateway platforms, highlighting the importance of secure integration and controlled transaction workflows.

In the proposed self-service printing station, QR codes are utilized within a controlled system environment to facilitate secure payment confirmation through a trusted payment gateway. This approach balances user convenience with system reliability while addressing potential risks associated with unverified or unauthorized QR code usage.

Common UNIX Printing System (CUPS)

The Common UNIX Printing System (CUPS) is widely implemented in Linux-based environments to manage print jobs, queues, and printer communication in automated systems. Its support for the Internet Printing Protocol (IPP) enables reliable handling of both local and networked printers, making it suitable for centralized print control applications (CUPS, 2023). Studies indicate that CUPS provides a stable and standardized framework for integrating printers with computing platforms that require continuous and unattended operation.

In automated service systems, CUPS facilitates efficient print workflow management by processing print requests, converting document formats, and transmitting data to printers without manual intervention. Its modular architecture allows seamless integration with application software and user interfaces, supporting real-time print job monitoring and control. In the proposed self-service printing station, CUPS serves as the core printing service that ensures consistent print output, system reliability, and compatibility with the Linux-based controller used in the implementation (CUPS, 2023).

Xendit

Xendit is a digital payment gateway widely used in Southeast Asia to facilitate secure and efficient electronic transactions across multiple payment channels. Korostashevych (March 2025) emphasized that Xendit supports various payment methods, including e-wallets and bank transfers, through API-based integration that enables automated transaction processing and real-time payment verification. Its compliance with PCI DSS standards further ensures the security and integrity of financial data during digital transactions.

In automated service systems, payment gateways such as Xendit play a critical role in ensuring reliable transaction handling and system synchronization. The platform’s ability to generate payment requests, confirm transaction status, and log payment records enables seamless integration with application-level services. In the proposed self-service printing station, Xendit serves as the intermediary payment service that validates GCash transactions and ensures that printing processes are executed only after successful payment confirmation, thereby improving transaction accuracy, security, and operational efficiency.

Gcash

GCash is a mobile-based digital wallet widely used in the Philippines to facilitate secure, cashless transactions. Afable (May 2024) notes that GCash supports payments, money transfers, bill settlements, and mobile commerce through a single integrated platform, enabling users to perform financial transactions efficiently and reliably. The application's regulatory compliance and security features, such as Customer Protect and QR code verification, ensure transaction integrity and user trust.

In automated payment systems, digital wallets like GCash serve as intermediaries that validate and authorize financial transactions in real time. Their integration via API allows seamless communication between the wallet and application-level services, ensuring that downstream processes are executed only after successful payment confirmation. In the proposed self-service printing staton, GCash functions as the primary cashless payment option. Its role is to confirm and log user payments, enabling the system to proceed with printing requests only after verification, thereby enhancing operational efficiency, security, and transaction accuracy.

Self Service Technology

Self service technologies (SSTs) are digital systems that allow users to independently perform transactions or access services without direct employee assistance. Malpass (2018) emphasized that SSTs enhance operational efficiency by automating routine tasks while providing users with real time access to services and information. Key characteristics of effective SSTs include system integration, ease of use, and reliable transaction processing, which support seamless interaction between users and service platforms.

The practical value of SSTs across industries has been well documented. Verma, Schulze, and Upreti (2025) discussed SSTs as a transformative force in sectors such as retail, hospitality, healthcare, and banking, where technologies like self checkout kiosks, self check in systems, digital service portals, and automated payment platforms reduce waiting time and improve customer satisfaction. Their study highlighted that SST adoption modernizes service delivery by improving efficiency, minimizing human intervention, and enhancing user experience through smart and connected technologies.

In automated service environments, SSTs play a critical role in reducing manual processes and ensuring consistent service delivery. In the proposed self service printing station, SST principles enable users to upload documents, initiate printing tasks, and complete transactions independently. By integrating SSTs with digital payment services such as GCash and Xendit, the system ensures secure and verified transactions while improving user convenience, transaction accuracy, and overall operational efficiency within the library setting.

Modern Academic Library Innovation

Academic literature positions innovation as a core function of libraries seeking to meet evolving user needs (Scott and McNamee, 2015; Walter and Lankes, 2015). This innovation often involves the introduction of new technologies, such as makerspaces and digital hubs, despite concerns related to cost and system complexity (Lehnen, 2019). Successful implementation is guided by frameworks such as design thinking, which emphasizes iterative and user centered development (Clarke, 2020), as well as strategies that promote user participation and external partnerships (Yeh and Walter, 2017). The proposed self-service printing station is situated within this context, representing a service innovation that renews a core library function by improving user convenience and operational efficiency through automation and cashless payment. This directly responds to the documented need for libraries to adapt and enhance their service models.

Cashless Payment

Cashless payments refer to transactions executed without the use of physical currency, relying instead on electronic methods such as credit cards, debit cards, mobile wallets, bank transfers, and contactless technologies. Team (2025) emphasized that cashless payment systems improve transaction speed, convenience, and security through mechanisms such as encryption, tokenization, and biometric authentication, ensuring reliable and efficient payment processing.

Consumer adoption of cashless payments has been examined across different contexts. Świecka (2019) conducted a nationwide survey in Poland comparing cash and cashless payment usage and found that consumer financial knowledge and skills significantly influence the acceptance and frequency of cashless transactions. The study highlighted that although cash payments remain prevalent, increased exposure to secure and user friendly digital payment systems encourages higher levels of cashless payment adoption. These findings underscore the importance of system reliability, ease of use, and trust in promoting cashless transactions.

In automated service environments, cashless payment systems play a critical role in improving operational efficiency by reducing manual handling of transactions, minimizing processing delays, and maintaining accurate transaction records. In the proposed self service printing station, cashless payments serve as the primary method for executing transactions. Integration with platforms such as GCash and Xendit ensures that printing services are activated only after successful payment verification, enhancing transaction security, system reliability, and user convenience within the library setting.

Self-Service Technology (SST) in Educational Institutions

According to Bell (2022), in Moving to Mobile: Space as a Service in the Academic Library, self-service technologies (SSTs) have increasingly been adopted by educational institutions to improve service efficiency, reduce queues, and provide students with greater autonomy. One notable example discussed is the installation of laptop and battery vending kiosks at Charles Library at Temple University. These kiosks allow students to borrow computing devices in a self-service manner, enhancing access particularly for students who do not regularly bring laptops or whose devices may be temporarily unavailable. Another study by Wendy S. Wilmoth (2020), in Circulating Laptops in a Two-Year Academic Library: A Formative Assessment, examined a laptop loan program at Georgia Piedmont Technical College. The study evaluated student satisfaction, identified logistical challenges, and analyzed how circulation policies influenced laptop usage, providing valuable insights for improving library technology services.

In the Philippines, the study of Carretero, & Apolinario (2022), SSTs are also gaining ground in academic libraries. For instance, the STARBOOKS (Science and Technology Academic and Research-Based Openly-Operated Kiosks) program was evaluated based on user perception" (UP Diliman) that examined how users perceive and interact with these kiosks to access digital resources. Also, Bukidnon State University (BukSU), (2023) recently introduced a Self-Check Kiosk for its library, aiming to ease borrowing and returning books by allowing users to perform these tasks without assistance, which was reported to improve convenience for students and staff.

Cashless Payment Adaptation and User Acceptance in the Philippines

The adoption of cashless payment systems in the Philippines has accelerated rapidly, particularly following the COVID-19 pandemic. Recent studies indicate a significant shift in consumer behavior, with Visa's Consumer Payment

Attitudes Study reporting that 84% of Filipino consumers attempted cashless transactions in 2021, with mobile wallets emerging as the most preferred channel at 64% (Luz, 2022). This trend continued into 2022, with 82% of Filipinos attempting cashless payments and 50% carrying less physical cash as digital options became more accessible (Navarro, 2023). The mobile wallet GCash has been central to this transformation, with studies demonstrating high user satisfaction due to its convenience, security features, and customer service (Montenegro, Afana & Alico, 2022). Research applying the Technology Acceptance Model (TAM) further supports this trend, identifying perceived ease of use, perceived usefulness, and trust as key factors driving e-wallet adoption among younger demographics in the Philippines (Belmonte et al., 2024). This widespread acceptance and positive user perception provides a strong foundation for integrating GCash into self-service systems like printing station in academic environments.

Operational and Institutional Benefits of Cashless Systems

The implementation of cashless systems offers significant operational advantages across various sectors. Studies demonstrate that cashless payment adoption reduces costs associated with handling physical cash while increasing transaction speed and accuracy (Rahman et al., 2022). These benefits extend to institutional settings, where digital payments streamline processes and reduce administrative workload. In the Philippines, this trend is evident in public service integrations, such as GCash partnerships with local government units for tax payments and social welfare distribution, which have improved efficiency and reduced physical queues (Gabay, 2021; Susilo & Dizon, 2023). Within educational contexts, similar benefits can be realized through self-service station, where cashless transactions minimize manual payment handling, accelerate service delivery, and provide automated transaction records for monitoring and reporting. These operational efficiencies directly align with the goals of implementing a cashless printing station in a university library setting.

Kiosk-Based Automation and System Architecture

Self service kiosks have demonstrated significant potential in improving operational efficiency and user experience across various service industries. In the Philippine context, Borbon (2024) examined the transformation of self service kiosks in the fast food industry and found that their adoption was driven by the need to enhance customer satisfaction, reduce waiting times, and streamline operations. The study highlighted how fast food chains invested in intuitive interfaces, multilingual support, and seamless payment integration to address usability challenges and improve service delivery. These findings indicate that well designed system architecture and user centered interfaces are critical to the successful deployment of self service technologies.

Academic projects further illustrate how automation and system architecture contribute to kiosk effectiveness. Geneveve et al. (2024) developed a smart coin operated printing kiosk that integrated backend monitoring, real time SMS notifications, and interactive user features. Their system architecture included error detection mechanisms and usage monitoring, which improved accuracy, efficiency, and learnability in a campus environment. The study demonstrated that incorporating automated monitoring and notification systems enhances reliability and user trust in self service applications.

Similarly, Bulaclac et al. (2023) designed an information kiosk with log monitoring capabilities that recorded user access patterns and system usage data. Their architecture supported both information dissemination and performance tracking, enabling institutions to analyze usage behavior and system effectiveness. These studies collectively emphasize that effective self service system architecture should integrate user interaction components with backend monitoring and reporting features.

In relation to the proposed self service printing station, these findings support the importance of a system architecture that combines automation, secure payment processing, user friendly interfaces, and administrative monitoring. By adopting proven architectural practices from existing kiosk systems, the printing station can achieve improved usability, operational efficiency, and reliability within the library environment.

AI and IoT-Enhanced Self-Service Kiosks

Based on Peña et al. (2023), integrates IoT, AI, and remote monitoring/services in kiosk systems to enable automation and real-time control. “An AI-integrated IoT-based Self-Service Laundry Kiosk with Mobile Application” describes “KILAO”, a self-service laundry kiosk. The system architecture combines hardware (laundry machines), IoT sensors, a mobile app, queue management via AI, and supports various payment options (cash, cashless, card). It demonstrates how combining backend cloud connectivity, AI prediction of queue/wait times, and user interface through a mobile app enhances both operational efficiency and user convenience. This one focus more on UX vs architectural/performance expectations: it compares what customers expect of a kiosk system (in terms of responsiveness, reliability, interface delays) and what they actually experience. Though less heavy on hardware architecture, it highlights system performance metrics such as processing speed, error rate, interface design, and how they are integral to a good kiosk architecture (Zhari et al., 2023).

Cybersecurity in Cashless Systems

As cashless systems proliferate, cybersecurity threats become more prominent, affecting user trust and system integrity. One study, “A Study on Cyber Attacks and Vulnerabilities in Mobile Payment Applications” Rivers et al., (2020), analyzes common mobile payment platforms like Apple Pay, Google Pay, Samsung Pay; it identifies points of vulnerability in authentication, authorization, verification chains, and transaction processing, which surveys 50 fintech / bank mobile payment apps in both high and low technological-advancement countries. It evaluates vulnerabilities per the OWASP / CWE standards, including issues related to cryptography, certificate handling, and insecure communications (Archibong et al., 2024).

Data Privacy and Protection in FinTech and E-Wallet Services

Fintech, a combination of financial services and digital technology, has become an essential component of modern transaction systems. Jafar, Hemachandran, and El-Chaarani (2024) examined the holistic adoption of fintech across financial services, emphasizing its role in improving transaction security, processing speed, and customer experience. Their study highlighted that fintech adoption extends beyond technological implementation, requiring effective governance, regulatory frameworks, risk management, and ethical considerations to ensure sustainable and secure financial operations. Through case studies, the authors demonstrated how fintech solutions are integrated into daily transactions, including mobile payments and digital wallets, reinforcing their growing relevance in cashless environments.

Despite its advantages, fintech adoption faces challenges related to trust, security, and privacy. Npbetia (2021) identified data privacy as a significant concern among users of mobile banking and e-wallet applications, noting that apprehensions regarding data collection, storage, and processing can discourage full adoption of cashless services. This finding is supported by Blumenstock and Kohli (2023), who discussed big data privacy issues in emerging market fintech systems and emphasized the need for technical safeguards such as encryption and differential privacy, alongside regulatory oversight, to protect consumer data. These studies collectively indicate that successful fintech implementation depends not only on technological capability but also on user trust and data protection mechanisms. In the context of the proposed self service printing station, fintech enabled payment platforms such as GCash and Xendit must incorporate secure transaction processing and data privacy measures to ensure user confidence. Aligning system design with established fintech governance and privacy practices supports reliable cashless transactions and enhances user acceptance in institutional service environments.

User Experience (UX), Convenience, and Accessibility in Digital Transactions

Usability and user experience (UX) are critical considerations in the design and evaluation of systems intended for human interaction. Lewis and Sauro (2021) discussed usability as the degree to which a system enables users to complete tasks effectively, efficiently, and with minimal errors, while UX encompasses the broader subjective experiences, perceptions, and satisfaction users derive during system interaction. Their work emphasized that effective system design combines scientific principles, creative design, and practical implementation to ensure positive user experiences.

The study highlighted usability testing as a primary method for evaluating system performance, focusing on the identification, prioritization, and resolution of usability issues through iterative testing. Lewis and Sauro (2021) further emphasized user centered design and design thinking as essential approaches in developing systems that align with user needs. These approaches rely on continuous feedback and iteration to refine system interfaces and workflows, ensuring improved usability and user satisfaction over time.

In automated service environments, usability and UX directly influence user acceptance and continued system use. For the proposed self service printing station, applying usability testing and user centered design principles ensures that users can easily navigate document uploading, print preview, and payment confirmation processes. Prioritizing usability and UX supports efficient task completion, reduces user errors, and enhances overall convenience, contributing to the system’s effectiveness and reliability in a library setting.

Demographic Influences on E-Wallet Adoption and User Experience

Accessibility and the broader impact of digital wallets on different demographic groups also play a big role. In “Digital wallets usage in the Philippines: determining the impact of demographics on complementarity versus substitutability, Reyes and Tenedero, (2021), investigated whether people view e-wallets as complementary to bank accounts or as substitutes, and how age, gender, employment affect that. They found that demographics significantly influence how users adopt and integrate digital wallets into their lives, which has implications for design (making interfaces and functions accessible for different user groups). Moreover “Examining consumers attitude towards E-wallet utilization as a payment method using technology acceptance model: A perspective from young generations in Nueva Ecija, Philippines” looks at UX and perceived risks (like privacy, security), showing that ensuring clear feedback, trust signals, and reliable performance are important for user convenience and continued usage (Leyesa et al., 2024).

Self-Service and Automation Initiatives in Mindanao

In Mindanao, several academic institutions have started adopting self-service and automated systems to improve operational efficiency and user accessibility. For instance, Bukidnon State University (BukSU), (2023), introduced a self-check kiosk for library transactions, enabling students to borrow and return books independently without librarian assistance. This innovation was designed to reduce waiting time and promote digital literacy among library users. Similarly, a study conducted at the University of Mindanao in Davao City explored the use of self-service kiosks in fast-food restaurants, revealing that customers highly value systems that offer convenience, speed, and reliability in transactions. The study concluded that user satisfaction is strongly influenced by the simplicity of navigation and the integration of cashless payment features (Del, Rojo & Marie, 2024).

Acopiado et al. (2022) from UP Mindanao conducted a study on digital payment adoption during the COVID-19 pandemic. They found that adoption in Mindanao businesses was influenced by infrastructure, firm size, and digital literacy. Their study suggests that regions outside Metro Manila face unique barriers such as weaker internet infrastructure, which must be considered when implementing kiosks in libraries.

Conceptual Framework

This study was anchored on the Input–Process–Output (IPO) model, which served as the conceptual guide in developing the Self-Service Printing Station in

SKSU Isulan Library with Cashless Payment Integration via GCash. The framework illustrates how the different components of the system interact—from the required inputs, to the processes of design and development, and finally to the expected outputs and feedback.

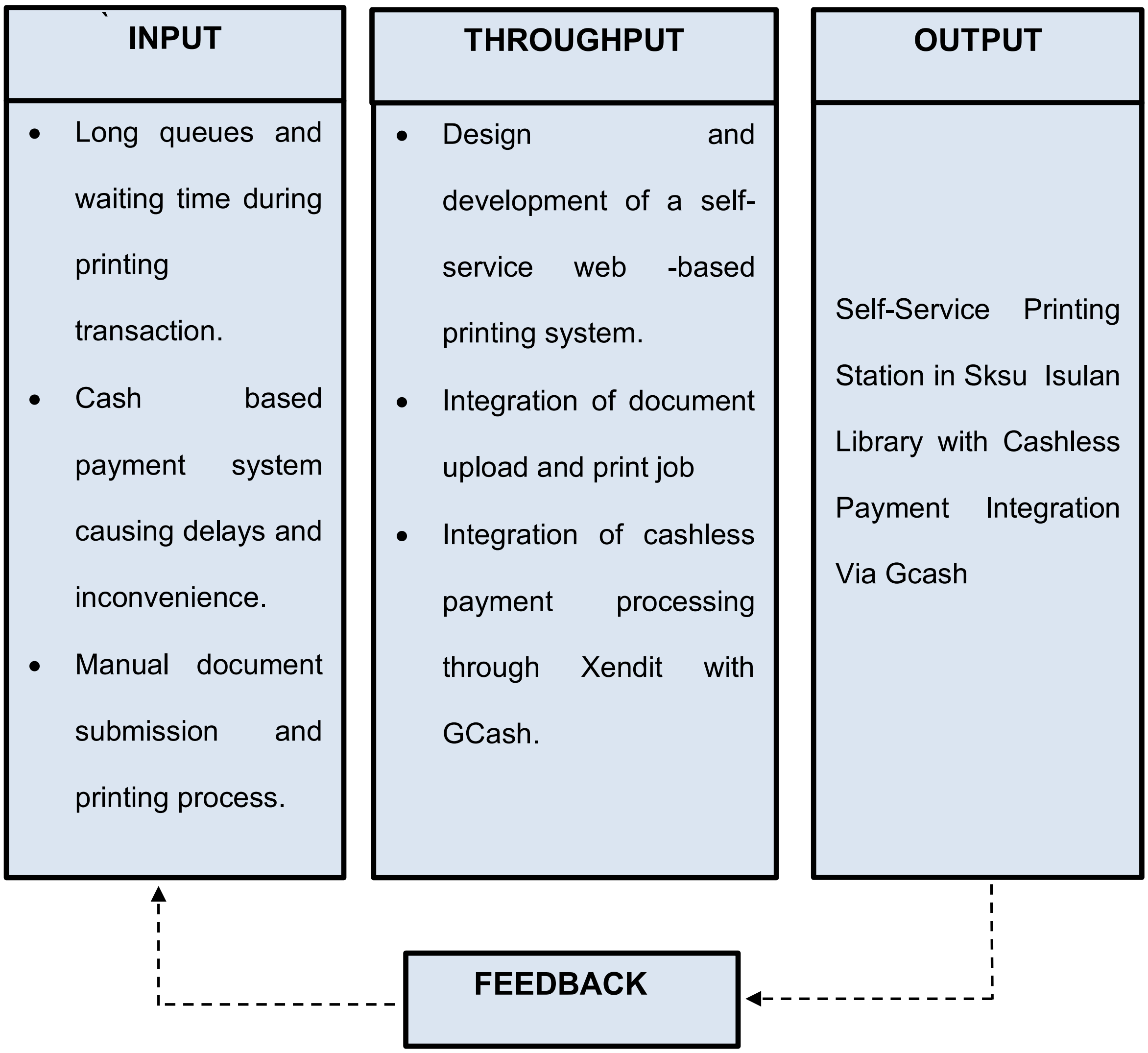

Figure 1. Conceptual Framework of the Study.

Input

The input phase focuses on the primary problems identified in the existing printing process at the SKSU Isulan Library. These include long queues and extended waiting time during printing transactions, manual document submission and printing procedures, and the use of cash-based payment methods that cause delays and inconvenience for users. These problems serve as the basis for determining the system requirements and defining the need for an automated self-service printing solution.

Process

These include long queues and extended waiting time during printing transactions, manual document submission and printing procedures, and the use of cash-based payment methods that cause delays and inconvenience for users. These problems serve as the basis for determining the system requirements and defining the need for an automated self-service printing solution.

Output

The output of the study is a fully functional Self Service Printing Station deployed at the SKSU Isulan Library. The system provides an automated and cashless printing service that allows users to upload documents, complete payments through GCash, and receive printed outputs without requiring staff assistance. This results in reduced queue time, faster transaction processing, and improved user convenience. It also generates weekly and monthly reports for the administrator to monitor sales and usage statistics.

Feedback

The feedback phase involves collecting and analyzing user evaluations to determine the system's effectiveness and reliability. Feedback from students, library staff, and administrators provides insights for improvement and serves as a basis for future enhancements of the self-service printing station, including interface design, system speed, and additional payment options.

CHAPTER III

METHODOLOGY

This chapter presents the Hardware Requirement used in the study, Bill of Supply and Materials, Project Duration, System Development Life Cycle, Context Diagram, Data Flow Diagram, Hardware Development and Methodology, and Perspective Plan.

Project Development Description

Tools and Equipment

These tools and equipment presented are the equipment used and needed for the completion of the project.

Table 1. Tools and Equipment

Quantity	Unit	Tool/Equipment
1	Pc.	Measuring tools
1	Pc.	Computer
1	Set	Screw drivers
1	Pc.	Electric Screw driver

Table 1 showed the device's tools and equipment used in constructing the Self-Service Printing Station in SKSU – Isulan Library with Cashless Payment Integration via GCash. The Measuring tools such as steel tape, rulers, and calipers were purposely used for accurate measurement of the system. Wires were essential in establishing electrical connections between the Raspberry Pi, inkjet printer, power supply, and other electronic modules. Screwdrivers, both flat and Philips types, were used for fastening and securing components such as the Raspberry Pi, power modules, and touchscreen monitor. These tools collectively ensured that the system was assembled accurately, functionally, and safely, resulting in a durable and reliable self-service printing station. Proper handling and selection of each tool contributed significantly to the efficiency and quality of the construction process.

Table 2. Bill of Supplies and Materials

Quantity	Unit	Item description	Unit Price (PHP)	Total (Php)
1	pc	Raspberry Pi 4 (4GB) with 32GB microSD card	4,000.00	4,000.00
1	pc	Official Raspberry Pi Power Supply (5V 3A)	700.00	700.00
1	pc	Touchscreen Monitor (10–15 inch)	4,000.00	4,000.00
1	pc	Multi-function Printer (Inkjet/Laser)	5,000.00	5,000.00
1	m	Ethernet Cable	500.00	500.00
1	Pc	WiFi Router	600.00	500.00
1	set	Device Enclosure (metal/plywood with acrylic)	1000	1,000.00
1	set	Hinges, Locks, Bolts, Nuts & Screws	500.00	500.00
1	set	Monitor Arm Stand	800.00	800.00
1	pc	Paper Tray	300.00	300.00
SUB TOTAL				**Php 16,400.00**
Contingency (10%)				**Php 1,640.00**
Grand Total				**Php 18,040.00**

Shown in Table 2 are the bills of supply and materials used in creating the Self-Service Printing Station in SKSU – Isulan Library with Cashless Payment Integration via GCash, including quantity, unit price, item description, and total cost per item, as well as the overall cost of the project design. The materials listed include both electronic components and construction supplies necessary for assembling the kiosk and integrating the cashless payment system. This detailed breakdown provides a clear overview of the resources required and aids in budgeting and procurement for the project. It also serves as a reference for planning future projects of similar nature, ensuring proper allocation of resources and cost efficiency.

Project duration

Figure 2. Project Duration

Activities	Duration (AY 2025-2026)									
	Aug	**Sept**	**Oct**	**Nov**	**Dec**	**Jan**	**Feb**	**Mar**	**Apr**	**May**
Researching for Title Recipient	→									
Gathering of Data	→									
Planning and Designing	→									
Title Defense		→								
Layout and Design		→	→							
Outline Defense			→							
Creating of the project				→	→					
Testing					→					
Evaluation of the Device					→					
Interpretation of Data						→				
Final Defense							→			
Finalization of Manuscript							→			
Publication of Journal								→		

Figure 2 shows the project duration of the Self-Service Printing Station in SKSU – Isulan Library with Cashless Payment Integration via GCash. As indicated, the conceptualization for the title proposal started in the first week of August 2025 until the end of the month. The title defense was conducted in the first week of September 2025. The gathering of tools and equipment took place in September 2025 until first week and second week of October, followed by the outline defense in October 2025.

Software Development Methodology

The software development methodology will adhere to the waterfall methodology model depicted below:

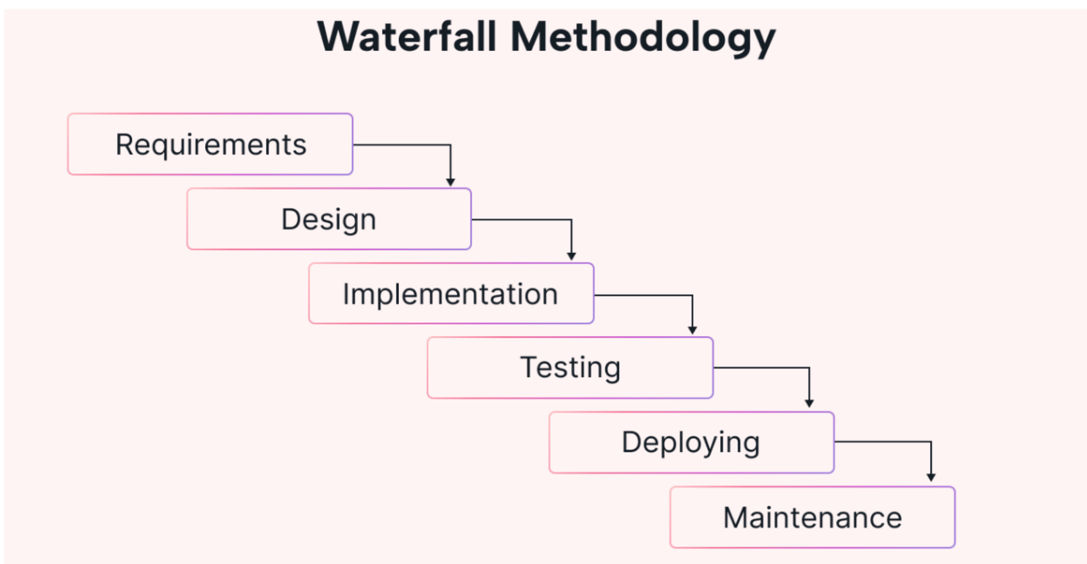

Figure 3. System Development Life Cycle

The researchers began by identifying the problems and challenges encountered by students in accessing affordable, efficient, and self-service printing facilities within the library. A project title was then formulated, together with the objectives, scope, and limitations of the study. Issues that arose during the design and development of the system were also recognized and addressed throughout the process of the study entitled Self Service Printing Station in SKSU – Isulan Library with Cashless Payment Integration via GCash.

The Waterfall Model was selected as the development approach since it follows a structured and linear framework with clearly defined phases. This model ensures that each stage of the development is completed before proceeding to the next, making it suitable for a system that requires well-documented processes and minimal iterations. The phases of the Waterfall Model applied in the study are presented as follows:

Requirements Analysis

The researchers gathered and analyzed the functional requirements of the system. This included file uploading, page counting, Xendit API, payment verification, and printing. Non-functional requirements, such as user-friendliness, reliability, and system security, were also identified and documented.

System Design

In the design phase, the hardware and software components were mapped and specified. The Raspberry Pi was chosen as the main controller, integrated with the printer, and touchscreen monitor. On the software side, CUPS was configured for printer management, Xendit API was designed for cashless transactions, and a database structure was outlined for transaction logging and reporting.

Implementation

The development phase involved programming the system interface using Python and web technologies, integrating the printer drivers, and linking the Xendit API. The Raspberry Pi was set up with the required operating system and software packages. Hardware components were also interfaced with the software to enable communication and automation.

Testing

Testing was conducted to evaluate the reliability and performance of the system. Each module was tested individually, such as file uploading, payment verification, and printing, followed by system-level testing. Trial-and-error procedures were carried out to identify bugs and errors, which were resolved before moving to the next stage.

Deployment and Maintenance

The finalized system was deployed in the SKSU – Isulan Library for actual use. During this stage, final adjustments were made to improve user interaction, transaction speed, and system security. Maintenance provisions were included to allow future updates, bug fixes, and integration of additional features if needed.

Context Diagram

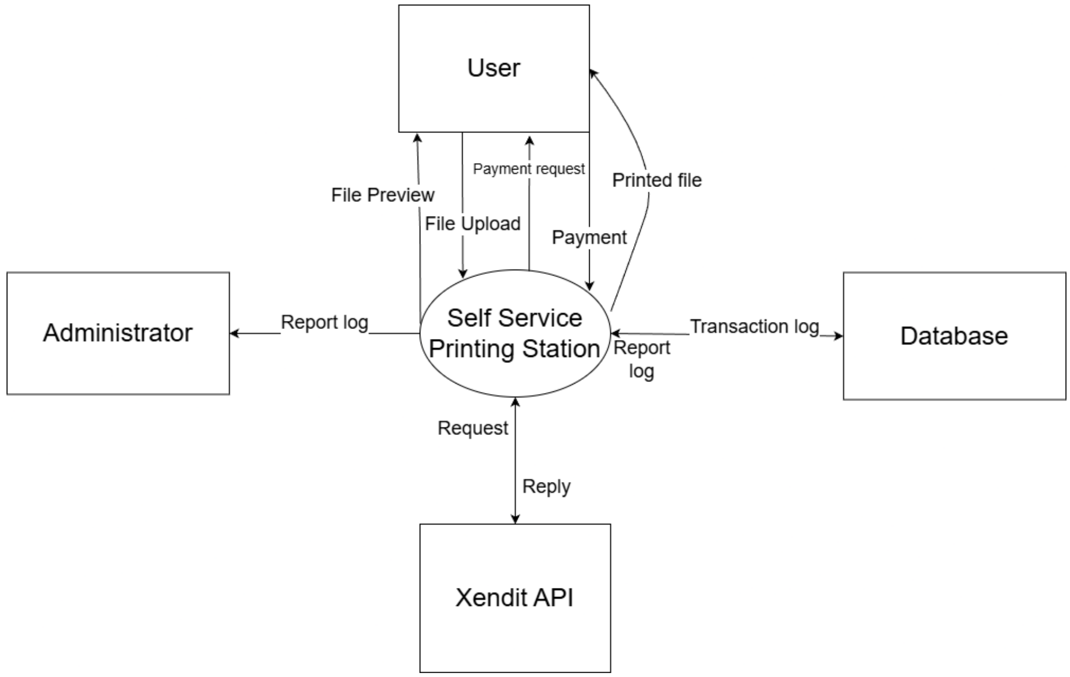

Figure 4. Context Diagram of the System

Figure 4 shows the context diagram of the Self-Service Printing Station in SKSU – Isulan Library with Cashless Payment Integration via GCash. It illustrates the system as a single process that interacts with external entities such as the User, Administrator, Xendit API, and the Database.

The diagram demonstrates the flow of inputs and outputs. The User uploads a file via qr code, previews the file, Xendit API for payment, generate qr code for payment, and receives the printed document as output. The Xendit API processes the payment request sent by the system and provides confirmation of successful transactions. The Administrator receives weekly and monthly log reports of sales and printed pages for monitoring purposes. The Database stores transaction records, log files, and payment information, while also sending feedback on completed printing jobs.

Data Flow Diagram

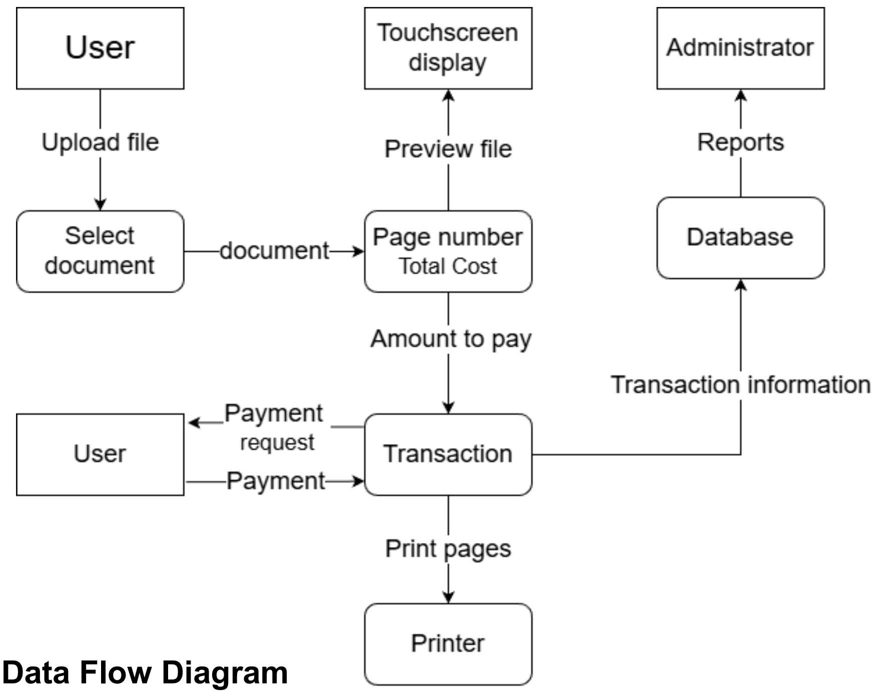

Figure 5. Data Flow Diagram

Figure 5 shows the Data Flow Diagram for the Self Service Printing Station in the SKSU Isulan Library, which integrates cashless payment via GCash. It illustrates how data moves between the system internal processes and external entities.

The User interacts with the touchscreen display to select and upload a document. The file is previewed before the system automatically counts the pages to determine the printing cost. The system then initiates a payment request, communicates with the GCash payment gateway, and generates a QR code or payment link for the user. Once the payment confirmation is received, the print job is sent to the Printer and the user collects the printed document.

All transaction information, including file details and payment status, is securely recorded in the Database. This stored data supports both real time transaction tracking and historical reporting. The Administrator can generate weekly and monthly sales reports and monitor system performance through an administrative dashboard.

The diagram effectively highlights how user inputs are transformed into physical and digital outputs, ensuring transparency throughout the printing and payment lifecycle. The process flow emphasizes the system efficiency and secure transaction handling from start to finish.

Entity Relationship Diagram

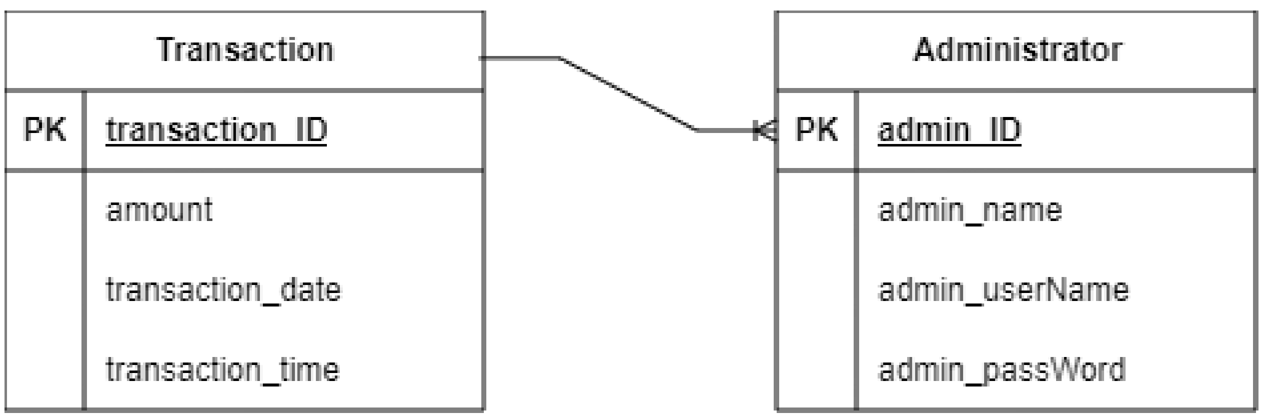

Figure 6. Entity Relationship Diagram

Figure 6 shows the Entity-Relationship Diagram (ERD) of the Self-Service Printing Station System. The diagram illustrates the two primary entities Administrator and Transaction and the relationship between them. The Administrator entity stores information about the authorized personnel responsible for monitoring and managing the system operations.

The Transaction entity records all printing activities performed on the system, including details such as the file page count, amount, transaction date, and transaction time. Each transaction is linked to a specific administrator, forming a one-to-many relationship, where one administrator can oversee multiple transactions.

This structure ensures organized data management within the system by maintaining clear accountability over printing activities and enabling efficient monitoring, reporting, and auditing of system usage.

Database Structure

Table 3. Administrator Table

Field Name	**Data type**	**Size**
admin_name	varchar	50
admin_id	Int	11
admin_userName	varchar	50
admin_passWord	varchar	50

Table 3 shows the admin account data structure of the system, including the admin’s information, such as name username, and password. The administrator account is stored in this table for storage purposes.

Table 4. Transaction Table

Field Name	Data Type	Size
transaction_id	INT	11
amount	INT	5
transaction_date	DATE	
transaction_time	TIME	

Table 4 shows the transaction history table, which stores records of all printing transactions made through the self-service printing station. Each transaction has a unique ID that helps identify and track specific printing activities. The table includes information such as the user's ID, total number of pages printed, total amount paid, and the corresponding date and time of the transaction. This table is essential for maintaining accurate records of printing usage, enabling efficient monitoring, management, and financial tracking of the system.

Hardware Development Methodology

The hardware development of the Self-Service Printing Station in SKSU – Isulan Library with Cashless Payment Integration via GCash will follow a systematic approach to ensure that all physical components are properly designed, constructed, and integrated with the software system. This methodology consists of three main parts: Perspective Plan, Construction Procedures, and Block Diagram. These components ensure that the design process is organized, functional, and aligned with the system requirements.

The Perspective Plan outlines the selection and arrangement of hardware components such as the processing unit, touchscreen display, printing device, and peripheral interfaces. The Construction Procedures describe the step by step assembly, wiring, and configuration of these components. Meanwhile, the Block Diagram provides a visual representation of the hardware architecture, illustrating the interaction between components and their connection to the software and payment modules.

Perspective Plan

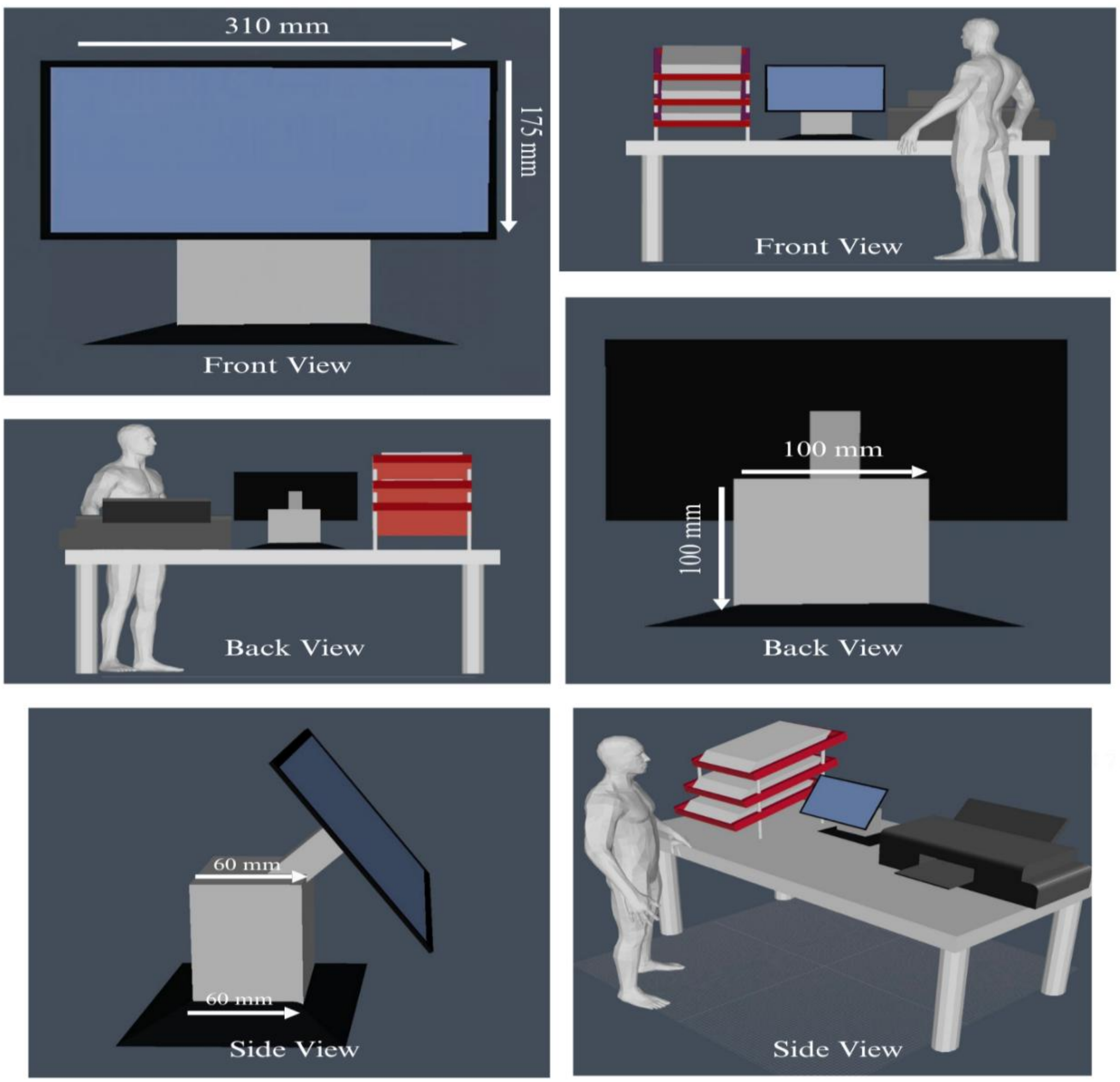

Figure 7. Perspective Plan of the Device

Figure 7 shows the Perspective Plan of the Self-Service Printing Station in SKSU – Isulan Library with Cashless Payment Integration via GCash, illustrating the physical layout and design of the system. The display used in the setup measures 14 inches with a 16:9 aspect ratio, equivalent to 310 mm in width and 175 mm in height. The Raspberry Pi unit is enclosed in a protective case measuring 100 mm in height, 100 mm in length, and 60 mm in width. This configuration provides a compact and organized arrangement, ensuring efficient interaction between the user interface, payment module, and printing components within the printing system.

Construction Procedures

The construction processes for developing the project were as follows:

1. Gather all materials, supplies, and equipment required for constructing the printing station, including the Raspberry Pi, touchscreen monitor, inkjet printer, power supply, wiring, cooling fan, enclosure materials, hinges, and locks.

2. Fabricate the device enclosure, which serves as the housing for all internal components such as the Raspberry Pi, power supply, and cable management system. Ventilation and cooling fans were also considered in the design to prevent overheating.

3. Install the main components, including the inkjet printer in the side section of the touchscreen monitor for document output, the touchscreen monitors at eye level for the user interface, the Raspberry Pi on its mounting bracket.
4. Program the Raspberry Pi with the appropriate code to manage the file upload, page counting, Xendit API, payment request handling, printer control, and database logging.
5. Connect the electrical and electronic components, including the Raspberry Pi GPIO pins, printer interface, and cooling system, ensuring proper circuit assembly and insulation with shrink tubes and cable ducts.
6. Conduct initial testing of all components to verify if the Raspberry Pi boots properly, the printer responds to print requests, the touchscreen displays the interface, the Xendit API detects payments, and the payment confirmation from GCash allows the print job to proceed.
7. Perform troubleshooting and final adjustments based on the test results. Retest the system to ensure complete functionality, safety, and reliability of the system before deployment in the SKSU – Isulan Library.

Flowchart

Figure 8 illustrates the flowchart of the functionality of the Self-Service Printing Station System. The process begins when the power system is activated, supplying energy to the Raspberry Pi, touchscreen monitor, and printer. Users upload their files through the interface, which are then received and processed by the Raspberry Pi. The user selects a file, and automatically counting the number of pages. The file is then displayed on the monitor for user verification.

If the user decides to proceed with printing, the system redirects to Gcash for payment. Once payment is confirmed, the printing process begins. However, if the user cancels the printing request or if the payment is not verified, the system terminates the process and returns to standby mode. The operation concludes when the printing is completed or the transaction is canceled, ready for the next user

.

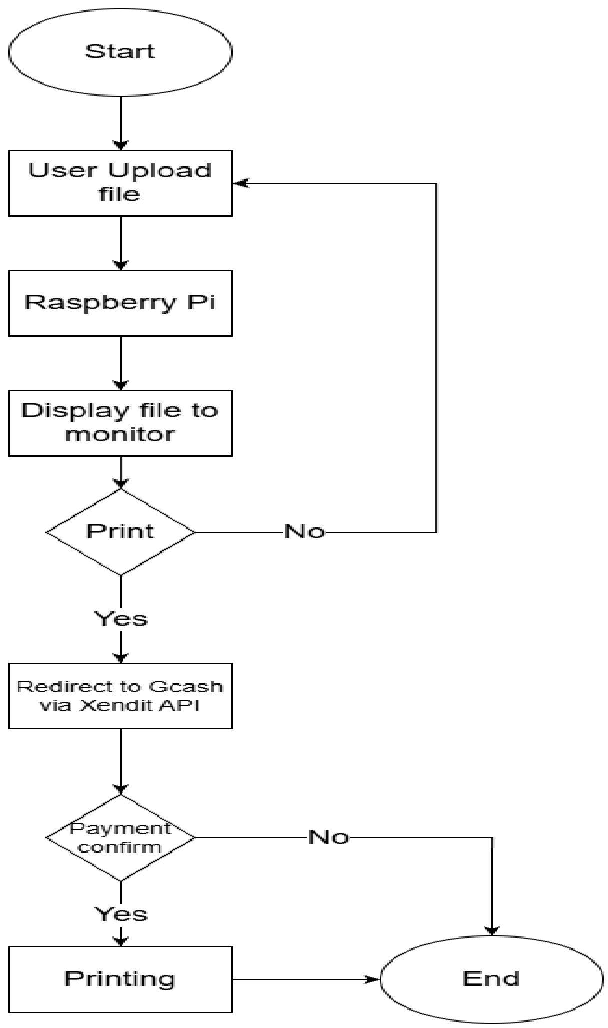

Figure 8. Flowchart of Self Service Printing Station

Evaluation Methodology

Research Design

The type of method that was used for this study was descriptive research. The researchers are required to gather information from the respondents by providing a questionnaire for evaluation of the study. The questions from the questionnaire evaluate the functionality, reliability and user convenience of the self-service printing Station.

Locale

Figure 9 shows the locale of the study, which will be conducted at Sultan Kudarat State University, Isulan Campus, specifically in the Isulan Library. The location was chosen because it is frequently visited by students who need to print their academic requirements, such as research papers, assignments, and reports. The library provides an ideal environment for implementing and testing the Self-Service Printing Station with Cashless Payment Integration via GCash, as it directly addresses the common issues of long queues, manual payment processes, and limited printing accessibility. Conducting the study in this setting allows researchers to evaluate the system's effectiveness in improving printing efficiency, promoting cashless transactions, and enhancing the overall convenience and productivity of students and library staff.

Figure 9. Satellite map of Sultan Kudarat State University Isulan Campus

Methods of Research

To fulfill the objectives of the study, a descriptive survey will be used. The creation of a study questionnaire is the first action taken by researchers. After the questionnaire has been created and distributed to respondents, the self-service printing station are evaluated. In the descriptive survey that was conducted, which will serve as the foundation for determining how reliable, effective and efficient the said system is, the respondents' perspectives, experiences, and views regarding the mobile application built are addressed.

Respondents of the Study

The respondents of this study will be composed of the first 200 users which would either be students from the College of Engineering, College of Computer Studies and College of Industrial Technology, faculty members, and staff of Sultan Kudarat State University – Isulan Campus, who are regular users of the university library and its facilities. A total of two hundred (200) respondents will be purposive sampling to participate in the evaluation of the developed self-service printing station. The inclusion of different groups of respondents ensured that the system was assessed from multiple perspectives — as end users, educators, and administrative personnel.

The respondents were chosen through random sampling to ensure fairness and representativeness of opinions from various departments within the university. The diversity of the respondents allowed for a comprehensive and balanced evaluation of the self-service printing kiosk, ensuring that the results of the study accurately reflect the system's overall performance and user satisfaction across different user groups.

Data Gathering Procedures

The researchers sought permission and authorization from the University faculty and staff. The researchers surveyed to gather the information that would be deemed helpful for the study. Throughout the survey, the researchers guaranteed that the respondents' identities were treated with the utmost confidentiality.

The following are the procedures on how the research gathers all the data and information regarding the project.

1. Prepare the web application and administrator website for evaluation.
2. Making research questionnaire.
3. Finalization of the questionnaire.
4. Distribution of questionnaire.
5. Testing and actual evaluation of the application.

Data Gathering Instruments

This study used a survey questionnaire as the data-gathering instrument. The instrument was chosen since it is cost-effective and ea sy to manage. The questionnaire comprises related questions wherein the respondents will rate the self-service printing station based on their experience. The researcher used a soft copy of the evaluation form to gather the data.

Table 5. Questions used in evaluating functionality of the system.

I. Functionality					
Questions	**Ratings**				
The SSPS system is functional in terms of:	**5**	**4**	**3**	**2**	**1**
1. The SSPS System boots without error.					
2. The system interface is responsive and user-friendly					
3. The file uploading operates smoothly.					
4. The GCash payment process functions properly.					
5. The system completes printing accurately.					

Table 6. Questions used in evaluating reliability of the system

II. Reliability					
Questions	**Ratings**				
The SSPS system is reliability in terms of:	**5**	**4**	**3**	**2**	**1**
1. The system consistently performs printing and payment transactions without failure.					
2. The SSPS responds promptly to user actions without delays or unresponsiveness					
3. The SSPS remains functional and accessible throughout library hours.					
4. Payments made through GCash are accurately verified and recorded.					
5. The SSPS provides accurate output according to the user's selected file.					

Table 7. Questins used in evaluating User Convenience of the system

III. User Convenience					
Questions	**Ratings**				
The SSPS system is user convenience in terms of:	**5**	**4**	**3**	**2**	**1**
1. The SSPS is easy to navigate and understand even for first-time users.					
2. The cashless payment option through GCash makes transactions faster and easier.					
3. The SSPS is accessible and available for use when needed.					
4. The SSPS allows me to complete printing tasks quickly and efficiently.					
5. Overall, the SSPS provides a convenient and satisfying user experience.					

Statistical Tools and Treatment of Data

After evaluating the Self-Service Printing Station in Sksu – Isulan Library with Cashless Payment Integration Via Gcash, the researchers tallied the evaluation results and collected the respondents' evaluation forms. The respondents' responses served as the basis for data tabulation.

Data analysis and interpretation were done using descriptive statistics such as the 5-point Likert Rating Scale. The 5-point Likert Rating Scale was statistically treated to determine the application's functionality, reliability, and user convenience. The 5-point Likert Scale will be computed using the formula below.

$$M = \frac{1(f) + 2(f) + 3(f) + 4(f) + 5(f)}{n}$$

M = mean

f = frequency

n = number of respondents

A. *The researchers used this formula in computing the standard deviation.*

$$\sigma = \sqrt{\frac{\sum(x-\mu)^2}{n}}$$

Where:

σ *= standard deviation*

$\sum$ *= is the sum of*

x = is a set of numbers

μ *= mean*

n = is the size of set

The data and responses of respondents in the questionnaires were collected according to the objectives of the designed project, and the results were interpreted by scale.

Table 8. Interpretative Scale used to Interpret the Mean

Numerical Scale	Interpretation
4.20 - 5.0	Strongly Agree
3.40 - 4.19	Agree
2.60 - 3.39	Moderately Agree
1.80 - 2.59	Disagree
1.00 - 1.79	Strongly Disagree

Table 8 displays the application's numerical ratings based on its overall mean. It demonstrates that the device Strongly Agree when its overall mean falls between 4.20 and 5.00. After that, the gadget receives descriptive ratings of Agree when its overall mean ranges from 3.40 to 4.19 and Moderately Agree when it ranges from 2.60 to 3.39. Additionally, the device will receive a descriptive rating of Disagree when its overall mean is between 1.80 and 2.59 and Strongly Disagree when its overall mean is between 1.00 and 1.79.

Table 9. Rating Scale used in Evaluating the Functionality of the System.

Rating	Interpretation
5	Highly functional
4	Functional
3	Moderately functional
2	Minimal functional
1	Not functional

Table 9 shows the rating scale and interpretation as a reference to evaluate the system functionality. A rating of 5 is considered highly functional, while a rating of 4 is functional. Similarly, a rating of 3 suggests moderately functional, a rating of 2 indicates less functional, and a rating of 1 indicates not functional.

Table 10. Rating Scale used in Evaluating the Reliable of the System.

Rating	Interpretation
5	Highly Reliable
4	Reliable
3	Moderately Reliable
2	Minimal Reliable
1	Unreliable

Table 10 presents the rating scale used to evaluate the *reliability* of the system. A rating of 5 indicates that the system is *highly reliable*, showing consistent performance without errors or failures. A rating of 4 means the system is *reliable*, functioning well under normal operating conditions. A rating of 3 suggests the system is *moderately reliable*, performing adequately but with minor issues. A rating of 2 reflects that the system is *less reliable*, experiencing occasional inconsistencies or errors, while a rating of 1 means the system is *not reliable* and frequently fails to perform its intended functions.

Table 11. Rating Scale used in Evaluating the User Convenience of the System.

Rating	Interpretation
5	Highly Convenient
4	Convenient
3	Moderately Convenient
2	Minimal Convenient
1	Inconvenient

Table 11 presents the rating scale used to evaluate the user convenience of the system. A rating of 5 indicates a highly convenient and user-friendly system, while 4 denotes a convenient system with minimal effort required. A rating of 3 reflects moderate convenience, whereas 2 indicates minimal convenience requiring additional effort. A rating of 1 signifies that the system is inconvenient and difficult to use.

Chapter IV
RESULTS and discussion

This chapter presents, analyzes, and interprets the data gathered from the respondents to evaluate the Self-Service Printing Station (SSPS) with Cashless Payment Integration via GCash. The evaluation is categorized into three main parameters: Functionality, Reliability, and User Convenience.

Results and Discussion

Presentation of the Developed System

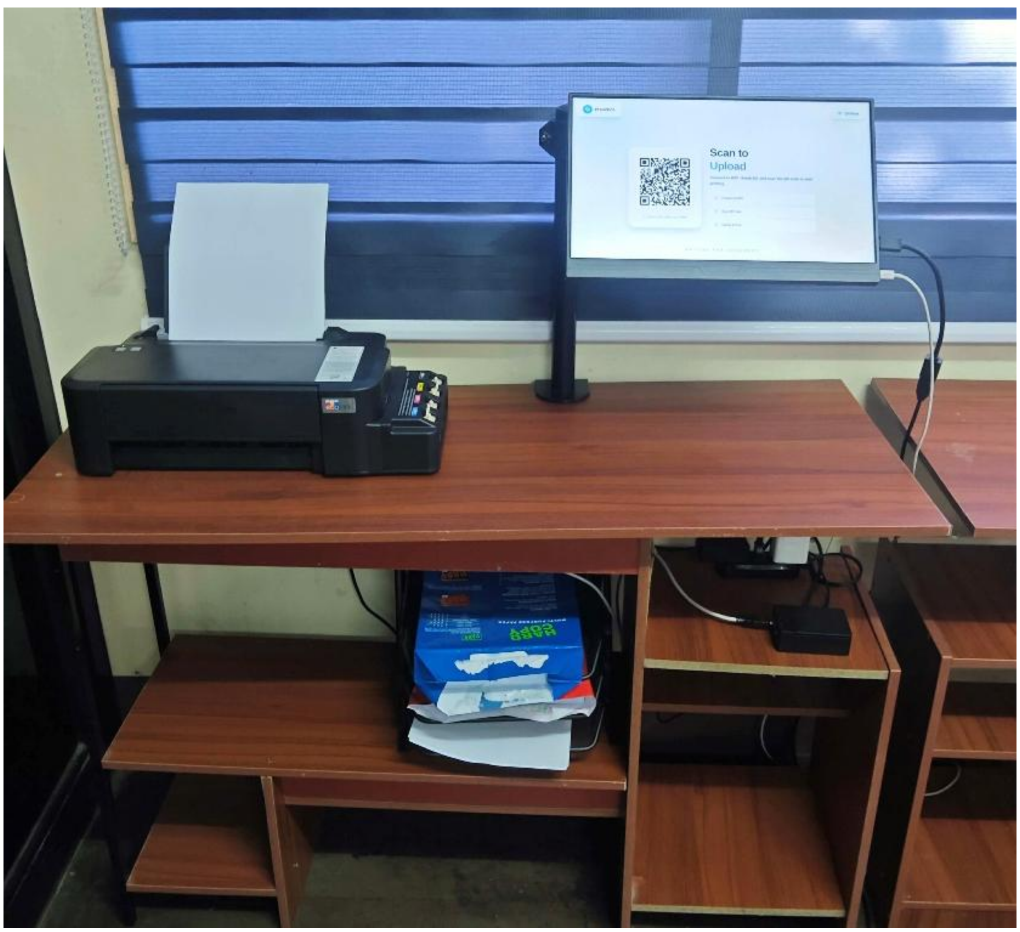

Figure 10. Actual Setup of SSPS

Figure 10 shows the design and development phase culminates in the fully operational Self-Service Printing Station (SSPS) prototype. Moving from initial 3D plans to physical construction, the final unit is organized into two main sections. The upper section contains an Epson L121 printer, and the display mount that supports the touchscreen display, and the lower tier holds the paper tray.

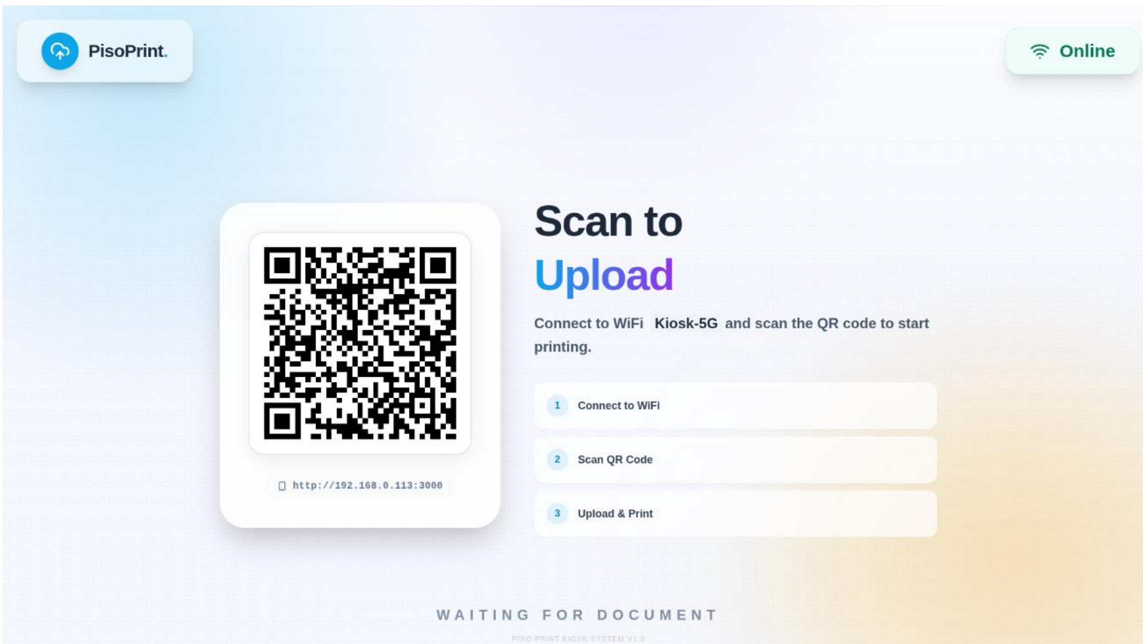

Figure 11. Landing Page of the SSPS

Figure 11 shows the system interface and contactless file upload mechanism of the Self-Service Printing Station (SSPS). The graphical user interface features a simple and intuitive design that guides users through a contactless file upload process. The main landing page serves as the entry point for both regular patrons and system administrators. To begin, users connect their mobile device to the SSPS's dedicated local network, establishing a direct connection to the kiosk's internal system without relying on external internet bandwidth. Once connected, the user scans the prominent QR code displayed on the screen, which opens a local web page on their phone where they can easily select and upload documents directly to the print queue. This method eliminates the need for USB drives and reduces physical contact with the machine. For system maintenance, authorized staff can access the backend by long-pressing the "PisoPrint" logo on the screen, a hidden gesture that reveals the administrator login page for monitoring transactions and Analytics.

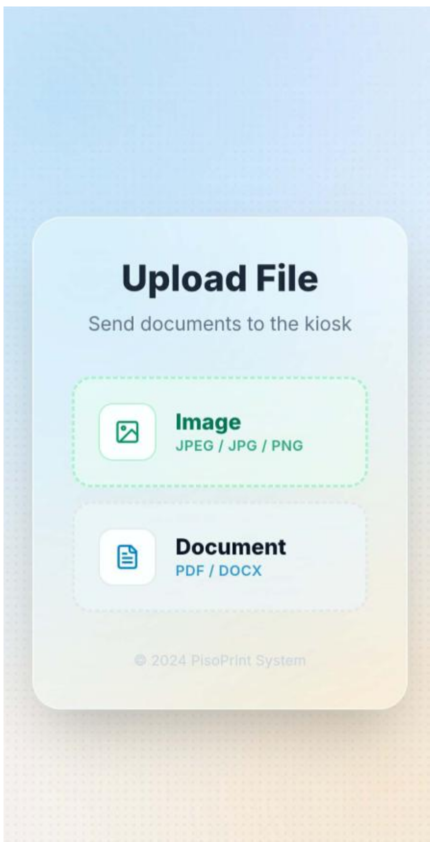

Figure 12. File Uploading Interface of SSPS

Figure 12 shows the file upload portal of the SSPS. After scanning the QR code, the user's mobile device displays this local web interface, which provides clear instructions for document submission. The portal supports common image formats including JPEG, JPG, and PNG, as well as document formats including PDF and DOCX. A simple upload button allows users to send their selected files directly to the kiosk print queue without creating an account or installing any application.

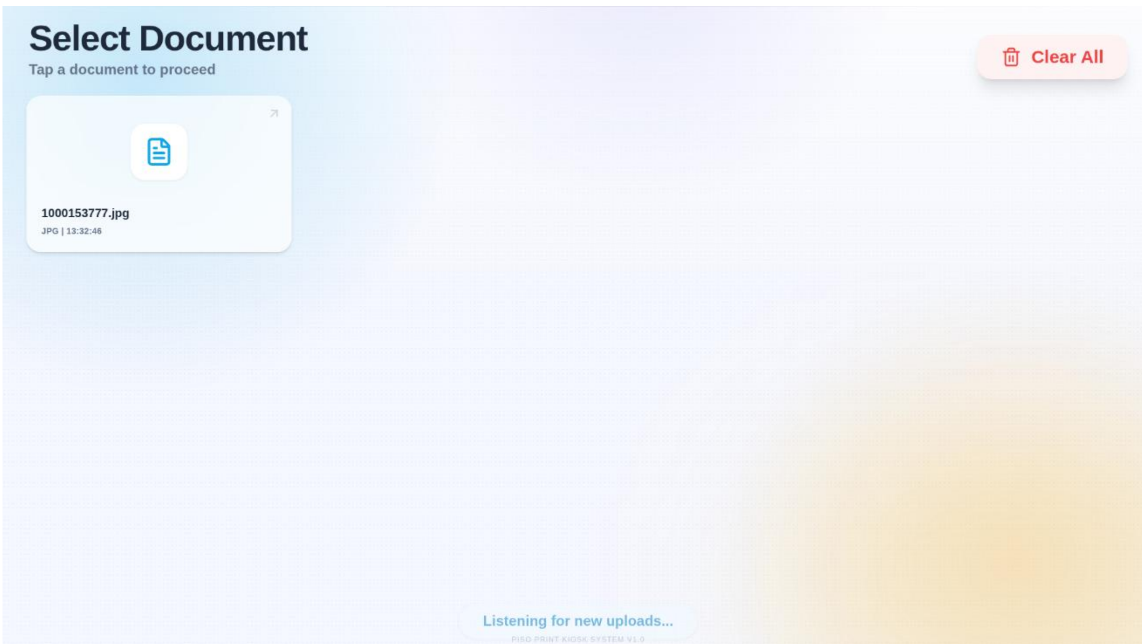

Figure 13. Document Selection of SSPS

Figure 13 shows the document selection screen of the SSPS mobile interface. Once a file has been successfully uploaded, the user is presented with a list of available documents. Each entry displays the file name, format type, and upload timestamp for easy identification. The user simply taps on the desired document to proceed with printing. The interface also remains active and displays a "Listening for new uploads..." status, allowing the user to send additional files to the queue without needing to rescan the QR code or refresh the page.

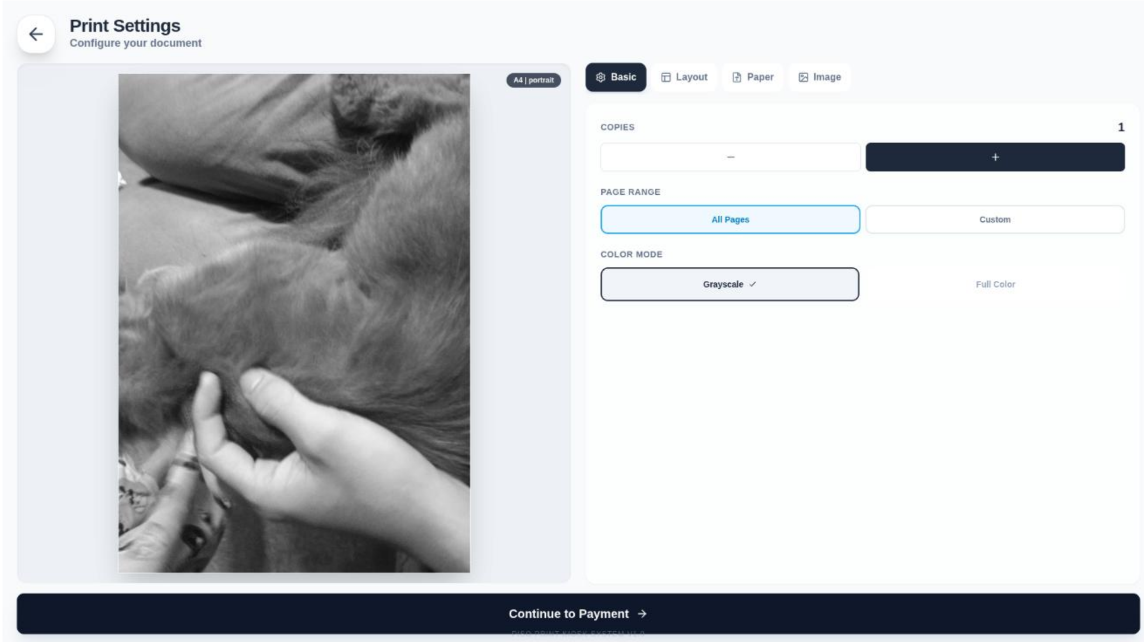

Figure 14. SSPS Print Configuration Interface: Basic Settings Screen

Figure 14 shows the basic print settings screen of the SSPS, which appears immediately after the user selects a document from the upload queue. This interface allows users to customize their print job before proceeding to payment. Common options available to all supported file types include the number of copies, page range selection (all pages or a custom range), color mode (grayscale or full color), as well as Layout and Paper type adjustments. If the selected file is a photo, an additional Image customization option becomes available, enabling further enhancements specific to image files. Once all desired settings are confirmed, the user taps the "Continue to Payment" button to advance to the final transaction step.

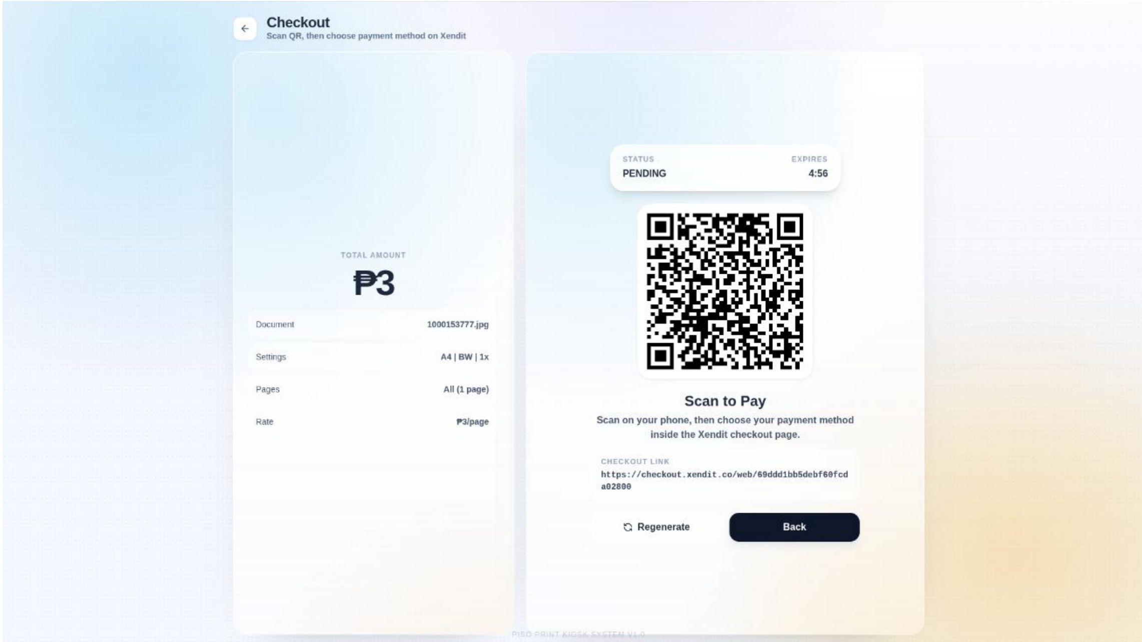

Figure 15. SSPS Payment Checkout Interface with Xendit QR Integration

Figure 15 shows the payment checkout screen of the SSPS, which appears after the user confirms print settings and taps "Continue to Payment." The interface displays a summary of the print job, including document name, selected settings, total number of pages, and the calculated amount due. A QR code generated through the Xendit API is presented for contactless payment. The user scans this QR code using their mobile device, which redirects them to a secure Xendit checkout page where they can select their preferred payment method. The transaction status is shown as pending until payment is completed, and a countdown timer indicates the session expiration time. A direct checkout link is also provided as an alternative access method.

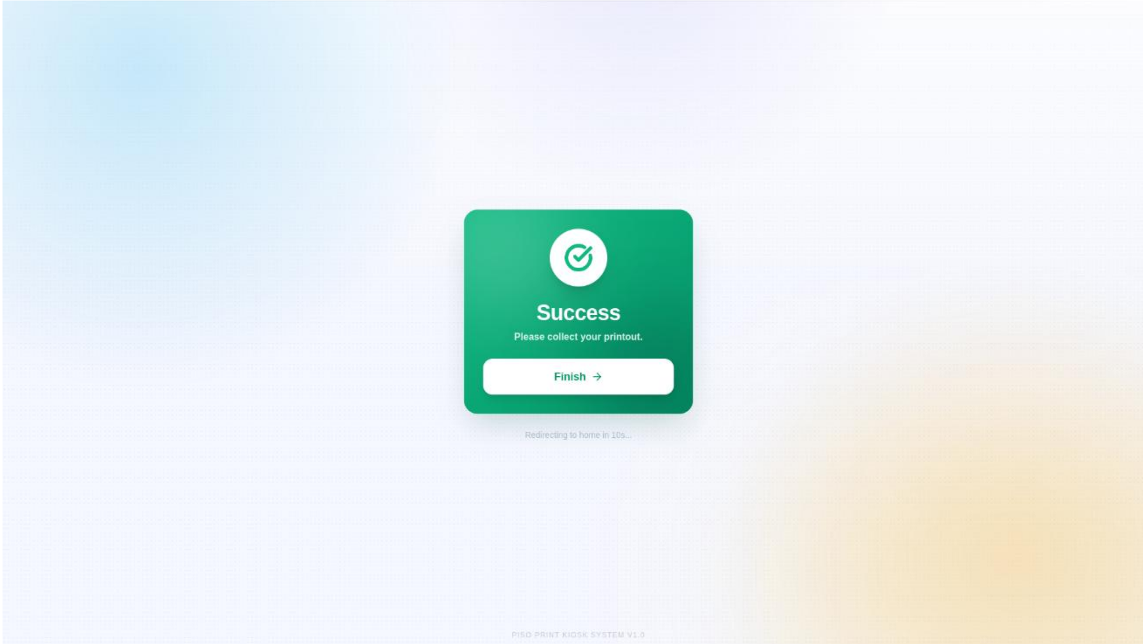

Figure 16. SSPS Payment Success and Print Completion Screen

Figure 16 shows the confirmation screen displayed upon successful payment and completion of the printing process. Once the Xendit transaction is verified, the system automatically releases the print job to the Epson printer. This final interface notifies the user that the transaction was successful. The user may tap the "Finish" button to immediately return the kiosk to the default landing page.

Figure 17. Administrative Authentication of SSPS.

Figure 17 shows the secure administrative access mechanism of the SSPS. The admin dashboard is intended solely for viewing system analytics and transaction data. By triggers a hidden menu by long-pressing the "PisoPrint" logo on the public touchscreen.

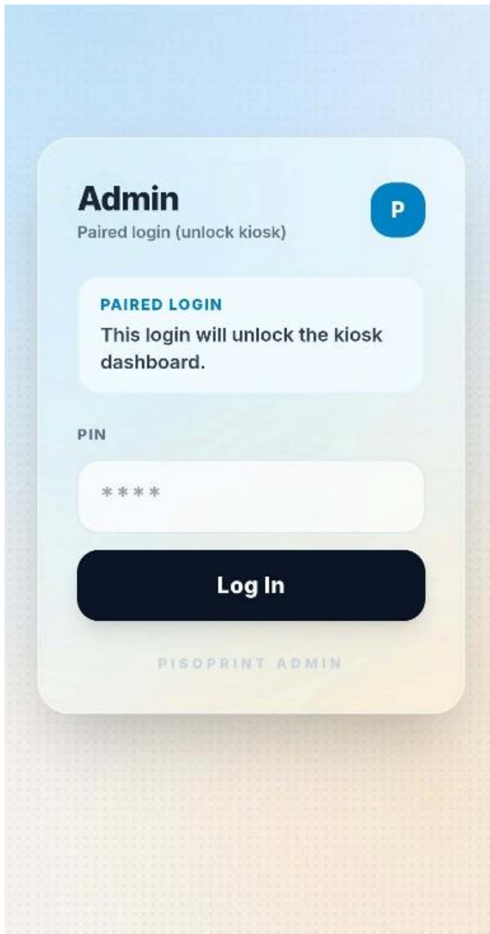

Figure 18. SSPS Mobile Admin Authentication: Paired Login Interface

Figure 18 shows the mobile admin login screen that appears after an authorized staff member scans the hidden Admin QR code displayed on the kiosk. This paired login interface requires the administrator to enter a secure PIN on their personal smartphone to authenticate and unlock the kiosk dashboard. This mobileexclusive credential entry method prevents sensitive login information from being exposed on the public touchscreen. Upon successful PIN verification, the kiosk display unlocks the administrative dashboard for system monitoring and analytics.

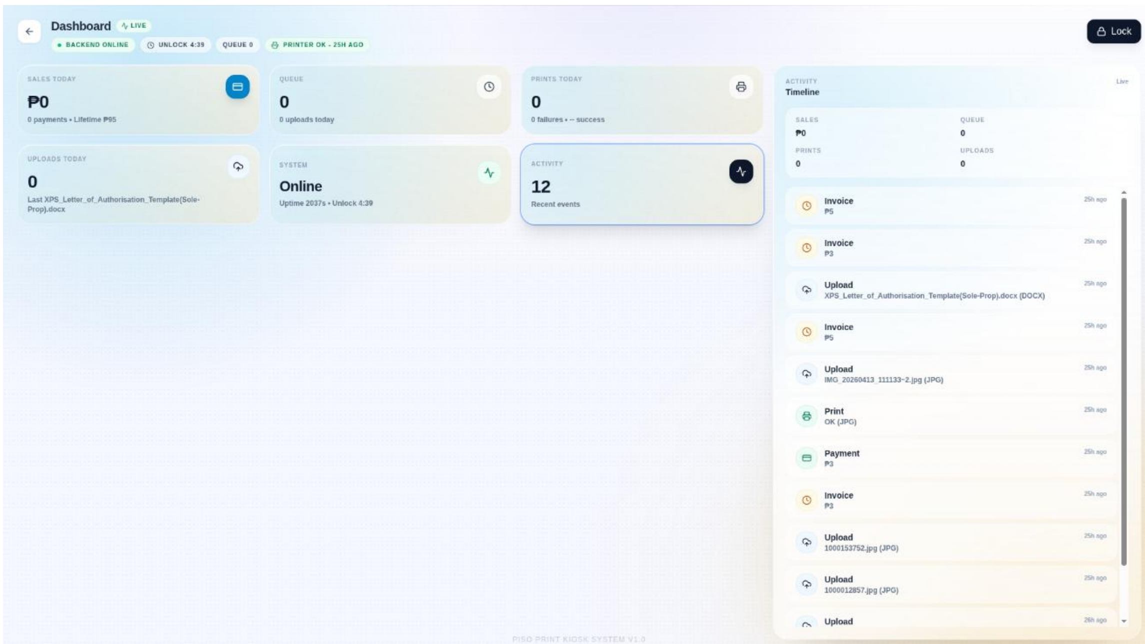

Figure 19. SSPS Administrative Dashboard: Analytics and Activity Monitor

Figure 19 shows the administrative dashboard of the SSPS, accessible only to authorized staff after secure mobile authentication. This interface provides real-time system analytics and transaction monitoring without allowing any direct control over printing operations. Key metrics displayed include backend online status, current print queue count, printer connectivity status, total sales for the day, lifetime earnings, and the number of successful uploads and prints. A detailed activity timeline logs all recent events such as uploads, payments, invoices, and print completions with timestamps for auditing purposes.

Evaluation of the Develop System

This section presents the results of the evaluation of the developed Self-Service Printing Station with Cashless Integration via GCash deployed at SKSU Isulan - Library. The evaluation focused on Functionality, Reliability, and User Convenience. Functionality assessed the system's ability to perform its intended operations, including file upload, print customization, and cashless payment processing. Reliability measured the consistency and stability of the system during actual use. User Convenience evaluated the overall ease and seamlessness of the contactless printing experience from the perspective of the end user. The responses of the participants were analyzed using the mean and interpreted based on the established rating scale.

Table 12. The Result ff Evaluation of the Functionality of the System

Question: The SSPS system is functionality in terms of:	**MEAN**	**STANDARD DEVIATION**	**INTERPRETATION**
The SSPS System boots without error.	4.58	0.513	SA
The system interface is responsive and user-friendly	4.65	0.493	SA
The file uploading operates smoothly.	4.605	0.503	SA
The GCash payment process functions properly.	4.605	0.489	SA
The system completes printing accurately.	4.625	0.494	SA

Table 12 presents the results on the functionality of the Self-Service Printing Station with Cashless Integration via GCash. The findings indicate that respondents evaluated the system's core operational capabilities as Strongly Agree (SA), with mean scores ranging from 4.58 to 4.65 across all five indicators. These consistently high ratings affirm that the developed system successfully meets its intended functional requirements, fulfilling the first evaluation objective (and validating the system's design as a web-based document handling and printing platform.

Respondents strongly agreed that the SSPS boots without error, as reflected by a mean of 4.58 (SD = 0.513), demonstrating stable initialization and operational readiness. The responsiveness and user-friendliness of the system interface received the highest mean score of 4.65 (SD = 0.493), underscoring the effectiveness of the graphical user interface design in guiding users through the printing workflow. The file uploading process was also rated favorably with a mean of 4.605 (SD = 0.503), confirming that the QR-based contactless upload mechanism operates smoothly and reliably.

In alignment with the objective to provide seamless cashless transactions, the GCash payment integration was evaluated with a mean of 4.605 (SD = 0.489), indicating that users found the payment process functional and straightforward. Finally, the accuracy of the printing output achieved a mean of 4.625 (SD = 0.494), confirming that the system correctly processes and produces documents according to user specifications.

The standard deviation values across all functionality indicators range narrowly from 0.489 to 0.513, signifying a high degree of consistency and consensus among the 200 respondents. Overall, the results substantiate that the SSPS is a fully functional, stable, and operates effectively.

Table 13. The Result ff Evaluation of the Convenience of the System

Question The SSPS system is reliability in terms of:	**MEAN**	**STANDARD DEVIATION**	**INTERPRETATION**
The system consistently performs printing and payment transactions without failure.	4.605	0.503	SA
The SSPS responds promptly to user actions without delays or unresponsiveness	4.635	0.492	SA
The SSPS remains functional and accessible throughout library hours.	4.625	0.494	SA
Payments made through GCash are accurately verified and recorded.	4.645	0.489	SA
The SSPS provides accurate output according to the user's selected file.	4.64	0.491	SA

Table 13 presents the results on the reliability of the self-service printing station with cashless integration via GCash. The findings indicate that respondents evaluated the system's consistency, stability, and dependability as strongly agree (SA), with mean scores ranging from 4.605 to 4.645 across all five indicators.

These consistently high ratings affirm that the developed system performs reliably under normal operating conditions, fulfilling the third evaluation and validating the system's capacity to deliver uninterrupted and accurate self-service printing.

Respondents strongly agreed that the system consistently performs printing and payment transactions without failure, as reflected by a mean of 4.605 (SD = 0.503). This finding demonstrates the robustness of the integrated hardware and software components in maintaining error-free operation during actual use. The responsiveness of the system to user inputs was also rated favorably, with a mean of 4.635 (SD = 0.492), indicating that the SSPS promptly executes commands without perceptible delays or unresponsiveness.

Furthermore, respondents affirmed the system's sustained availability, achieving a mean of 4.625 (SD = 0.494). This suggests that the kiosk remains functional and accessible throughout the library's operating hours, reinforcing its dependability as a public service utility. The accuracy of GCash payment verification and record-keeping was evaluated with a mean of 4.645 (SD = 0.489), indicating that users trust the system to correctly process and log cashless transactions. Finally, the accuracy of the printed output relative to the user's selected file received a mean score of 4.64 (SD = 0.491), confirming that the system reliably reproduces documents without errors or discrepancies. The standard deviation values across all reliability indicators range narrowly from 0.489 to 0.503, signifying a high degree of consistency and consensus among the 200 respondents. The low variability reinforces the validity of the findings and indicates a shared perception that the ssps operates as a stable and trustworthy self-service solution. Overall, the results substantiate that the developed system meets the required standards of reliability necessary for continuous deployment in a public library setting.

Table 14. The Result ff Evaluation of the Convenience of the System

Questions The SSPS system is user convenience in terms of:	**MEAN**	**STANDARD DEVIATION**	**INTERPRETATION**
The SSPS is easy to navigate and understand even for first-time users.	4.605	0.503	SA
The cashless payment option through GCash makes transactions faster and easier.	4.635	0.492	SA
The SSPS is accessible and available for use when needed.	4.625	0.494	SA
The SSPS allows me to complete printing tasks quickly and efficiently.	4.645	0.489	SA
Overall, the SSPS provides a convenient and satisfying user experience.	4.64	0.491	SA

Table 14 presents the results on the user convenience of the Self-Service Printing Station with Cashless Integration via GCash. The findings indicate that respondents evaluated the system's ease of use and overall accessibility as Strongly Agree (SA), with mean scores ranging from 4.615 to 4.675 across all five indicators. These consistently high ratings affirm that the developed system delivers a seamless and user-friendly printing experience, fulfilling the fourth evaluation objective (4.3 User Convenience) and validating the design of the SSPS as an accessible and efficient self-service station.

Respondents strongly agreed that the system is easy to navigate and understand even for first-time users, as reflected by a mean of 4.615 (SD = 0.498). This finding demonstrates that the intuitive layout of the graphical user interface effectively minimizes confusion and supports independent use without prior instruction. The cashless payment option through GCash was rated as making transactions faster and easier, with a mean of 4.655 (SD = 0.486), indicating strong approval of the contactless payment integration as a time-saving feature that enhances the overall workflow.

Furthermore, respondents affirmed the system's accessibility and availability, achieving a mean of 4.635 (SD = 0.492). This suggests that the kiosk is perceived as reliably present and ready for use whenever needed, which is essential for a self-service station in a public library environment. The efficiency of the SSPS in completing printing tasks was evaluated with a mean of 4.66 (SD = 0.484), confirming that users find the process quick and streamlined. Finally, the overall user experience received the highest mean score of 4.675 (SD = 0.476), reflecting an exceptionally positive and satisfying perception of the system's convenience.

The standard deviation values across all convenience indicators range narrowly from 0.476 to 0.498, indicating a strong consensus among the 200 respondents. The low variability reinforces the reliability of the findings and underscores a shared positive perception of the SSPS as a convenient, efficient, and user-centered printing solution. Overall, the results substantiate that the system successfully meets the convenience requirements of its intended users and contributes to a frictionless self-service experience.

Table N. The Summary Result of Evaluation of the System

EVALUATION OF THE SYSTEM	MEAN	INTERPRETATION
Functionality	4.621	Highly functional
Reliability	4.63	Highly Reliable
Convenience	4.648	Highly Convenient
Overall Rating	**4.633**	**Strongly Agree**

Table [N] presents the overall evaluation of the Self-Service Printing Station with Cashless Integration via GCash based on three core performance dimensions: functionality, reliability, and user convenience. The overall mean rating of 4.633 is interpreted as "Strongly Agree," indicating that the 200 respondents held a highly positive and favorable assessment of the developed system.

Among the three evaluation criteria, User Convenience received the highest mean score of 4.648, interpreted as "Strongly Agree." This demonstrates that users found the system exceptionally easy to navigate, efficient in completing tasks, and highly accessible. The cashless GCash integration notably contributed to a faster and more seamless transaction experience. Reliability followed with a mean of 4.630, also interpreted as "Strongly Agree," signifying that the system consistently performed printing and payment operations without failure, responded promptly to user inputs, and remained functional throughout library hours. Lastly, Functionality achieved a mean of 4.621, interpreted as "Strongly Agree," confirming that all core features—including system boot-up, file uploading, payment processing, and accurate printing—operated as intended.

The consistently high ratings across all three dimensions, with mean values ranging narrowly from 4.621 to 4.648, affirm that the SSPS successfully meets its design objectives. The system provides a fully functional, dependable, and user-friendly self-service printing solution well-suited for deployment in public library environments.

Chapter V
Summary, conclusion, and Recommendation

This chapter presents the summary of the development process, the conclusions derived from the research, and recommendations for the future enhancement of the system

Summary of Findings

The findings of the study revealed several insights regarding the performance and usability of the Self-Service Printing Station with Cashless Integration via GCash. The system was successfully deployed as a fully operational kiosk integrating an Epson multifunction printer, a Raspberry Pi-based control unit, and a web-based user interface for contactless document handling and cashless payment processing.

The evaluation of the system focused on three core dimensions: Functionality, Reliability, and User Convenience, utilizing responses from 200 library patrons. The overall mean rating of 4.633, interpreted as "Strongly Agree," indicates a highly favorable assessment of the system across all criteria.

In terms of Functionality (Mean = 4.621), respondents strongly agreed that the SSPS boots without error, features a responsive and user-friendly interface, facilitates smooth file uploading, processes GCash payments properly, and completes printing tasks accurately. The system effectively fulfills its intended operational requirements as a self-service document handling and printing platform.

Regarding Reliability (Mean = 4.630), users affirmed that the system consistently performs printing and payment transactions without failure, responds promptly to user actions, remains functional and accessible throughout library hours, accurately verifies and records GCash payments, and produces printed outputs that precisely match the user's selected files. The low standard deviation values observed across reliability indicators demonstrate a strong consensus regarding the system's stability and dependability.

With respect to User Convenience (Mean = 4.648), the highest-rated dimension, respondents emphasized that the SSPS is easy to navigate and understand even for first-time users, and that the cashless payment option through GCash significantly accelerates transactions. The system was perceived as accessible when needed, efficient in completing printing tasks, and overall provides a convenient and satisfying user experience. The intuitive graphical user interface, QR-based contactless file upload mechanism, and seamless GCash integration collectively contributed to a frictionless self-service workflow.

However, certain limitations were observed during the development and evaluation phases. The administrative dashboard is designed exclusively for viewing system analytics and transaction logs, with no capability for remote configuration or direct control over printing operations. Additionally, the system enforces a 10-second inactivity timeout on the confirmation screen to automatically return the kiosk to the landing page, which, while promoting security and readiness for the next user, may occasionally interrupt users who delay collecting their printouts. Furthermore, the system's reliance on a dedicated local network for file transfer requires users to manually connect their mobile devices to the SSPS hotspot, which may present a minor learning curve for less tech-savvy patrons.

Overall, these findings indicate that the SSPS is a highly functional, reliable, and convenient self-service solution that successfully addresses the need for contactless, cashless printing in a public library setting. Minor areas for enhancement were identified to further optimize the user experience and administrative capabilities.

Conclusion

The Self-Service Printing Station with Cashless Integration via GCash was successfully developed as a web-based, microcontroller-driven kiosk that integrates an Epson multifunction printer, a Raspberry Pi control system, and the Xendit API for GCash payment processing. The system effectively enables users to upload documents via a QR-based local web portal, customize print settings, complete cashless transactions, and retrieve printed outputs without requiring staff assistance or physical contact with shared hardware.

The evaluation based on three core performance dimensions—Functionality, Reliability, and User Convenience—revealed that the system is rated as "Strongly Agree" across all indicators. Respondents recognized its ability to perform core operations without error, maintain consistent and stable performance, and deliver a seamless and user-friendly printing experience. The strong agreement among the 200 respondents, reflected by low standard deviation values, further validates the system's acceptability, practicality, and readiness for real-world deployment.

The successful implementation of the SSPS demonstrates that contactless, cashless self-service solutions can be effectively deployed in academic and public library environments. The system eliminates the security risks and hardware degradation associated with USB flash drives, reduces staff workload by automating the printing and payment workflow, and provides library administrators with valuable analytics for monitoring usage patterns and transaction history.

Recommendation

Based on the findings and conclusions of the study, the following recommendations are proposed for the future enhancement and broader implementation of the Self-Service Printing Station:

1. Enhancement of Administrative Capabilities. Future iterations of the system should expand the administrative dashboard to include remote configuration options, such as adjusting printing rates, managing file format restrictions, and performing system diagnostics without requiring physical access to the sytem. This would streamline maintenance and allow library staff to respond more efficiently to changing operational needs.

2. Integration of Multiple Payment Gateways. While the current GCash integration via Xendit provides a seamless cashless experience, incorporating additional payment options—such as Maya, bank transfers, or card payments—would increase accessibility and accommodate a wider range of user preferences.

3. Implementation of Cloud-Based File Transfer. To eliminate the need for users to manually connect to the system local network, future versions could leverage cloud-based file upload solutions, such as a temporary web-based dropbox or integration with institutional email or learning management systems. This would further reduce friction and enhance user convenience.

3. Improved Session Management and User Feedback. The 10-second inactivity timeout on the confirmation screen could be made configurable or replaced with a more user-friendly prompt, allowing users additional time to retrieve their printouts before the session expires. Additionally, incorporating audio or visual alerts upon print completion would improve accessibility and user awareness.

4. Longitudinal Usage Study. A follow-up study is recommended to assess the long-term reliability and user satisfaction of the SSPS over an extended deployment period. Such a study would provide valuable data on hardware durability, maintenance requirements, and evolving user expectations in a real-world library environment.

5. Deployment in Additional Locations. Given the positive evaluation results, the SSPS is recommended for deployment in other libraries, co-working spaces, and educational institutions seeking to modernize their printing services with contactless and cashless capabilities.

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THESIS

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APPENDICES

Appendix 1

Form 1

Republic of the Philippines
SULTAN KUDARAT STATE UNIVERSITY
College of Engineering
Isulan Campus
Isulan, Sultan Kudarat

PLAN OF COURSE WORK

Name: **NATHANIEL C. LAGNAODA** Course: **BSCpE**
Major Course:

Course No.	**Course Description**	**Unit**	**Remarks**
AC211	Fundamentals of Electrical Circuits	4.00	Passed
AC221	Fundamentals of Electronic Circuits	4.00	Passed
BES211	Computer-Aided Drafting	1.00	Passed
BES212	Engineering Economics	3.00	Passed
BES214	Engineering Management	2.00	Passed
BES321	Technopreneurship 101	3.00	Passed
CpE111	Computer Engineering as a Discipline	1.00	Passed
CpE112	Programming Logic and Design	2.00	Passed
CpE121	Object Oriented Programming	2.00	Passed
CpE122	Discrete Mathematics	3.00	Passed
CpE211	Data Structures and Algorithms	2.00	Passed
CpE221	Numerical Methods	3.00	Passed
CpE222	Software Design	4.00	Passed
CpE311	Logic Circuits and Design	4.00	Passed
CpE312	Operating Systems	3.00	Passed
CpE313	Data and Digital Communications	3.00	Passed
CpE314	Introduction to HDL	1.00	Passed
CpE315	Feedback and Control Systems	3.00	Passed
CpE316	Fundamentals of Mixed Signals and Sensors	3.00	Passed
CpE317	Computer Engineering Drafting and Design	1.00	Passed
CpE321	Basic Occupational Health and Safety	3.00	Passed
CpE322	Computer Networks and Security	4.00	Passed
CpE323	Microprocessors	4.00	Passed
CpE324	Methods of Research	3.00	Passed
CpE325	CpE Laws and Professional Practice	2.00	Passed
CpE413	Emerging Technologies in CpE	3.00	Passed
CpE423	On the Job Training	3.00	Passed
CpEELEC311	Cognate / Elective Course 1**	3.00	Passed
CpEELEC321	Cognate / Elective Course 2**	3.00	Passed
CpEELEC411	Cognate / Elective Course 3**	3.00	Passed
EnE111	Environmental Science and Engineering	3.00	Passed

Republic of the Philippines
SULTAN KUDARAT STATE UNIVERSITY
College of Engineering
Isulan Campus
Isulan, Sultan Kudarat

GE701	Mathematics in the Modern World	3.00	Passed
GE702	Purposive Communication	3.00	Passed
GE703	Ethics	3.00	Passed
GE704	Science, Technology and Society	3.00	Passed
GE705	The Contemporary World	3.00	Passed
GE706	Art Appreciation	3.00	Passed
GE707	Readings in Philippine History	3.00	Passed
GE708	Understanding the Self	3.00	Passed
GE709	The Life and Works of Jose Rizal	3.00	Passed
GE711	Culture of Mindanao	3.00	Passed
GE712	Gender and Society	3.00	Passed
GE713	Kontekstwalisadong Komunikasyon sa Filipino (KOMFIL)	3.00	Passed
GE714	Sosyedad at Literatura / Panitikang Panlipunan (SOSLIT)	3.00	Passed
GE715	Filipino sa Iba't ibang Disiplina (FILDIS)	3.00	Passed
MATH011	Calculus 1	3.00	Passed
MATH012	Calculus 2	3.00	Passed
MATH201	Differential Equations	3.00	Passed
MATH202	Engineering Data Analysis	3.00	Passed
NSTP101	National Service Training Program 1	3.00	Passed
NSTP102	National Service Training Program 2	3.00	Passed
PE101	Physical Fitness and Self-Testing Activities	2.00	Passed
PE102	Rhythmic Activities	2.00	Passed
PE103	Recreational Activities	2.00	Passed
PE104	Team Sports	2.00	Passed
SCI111	Chemistry for Engineers	4.00	Passed
SCI121	Physics for Engineers	4.00	Passed

Total Number of Units Required for the Period : **175 UNITS**
Total Numbers of Units Earned : **159 UNITS**

Certified Correct:

Approved:

JOVANIE S. LINGCAY
Campus Registrar

____
Date

ROMMEL M. LAGUMEN
Campus Director

____
Date

Appendix 2

Form 2

Republic of the Philippines
SULTAN KUDARAT STATE UNIVERSITY
College of Engineering
Isulan Campus
Isulan, Sultan Kudarat

PLAN OF COURSE WORK

Name: **GEORGE ANDRIE D. GATAN** Course: **BSCpE**
Major Course:

Course No.	Course Description	Unit	Remarks
AC211	Fundamentals of Electrical Circuits	4.00	Passed
AC221	Fundamentals of Electronic Circuits	4.00	Passed
BES211	Computer-Aided Drafting	1.00	Passed
BES212	Engineering Economics	3.00	Passed
BES214	Engineering Management	2.00	Passed
BES321	Technopreneurship 101	3.00	Passed
CpE111	Computer Engineering as a Discipline	1.00	Passed
CpE112	Programming Logic and Design	2.00	Passed
CpE121	Object Oriented Programming	2.00	Passed
CpE122	Discrete Mathematics	3.00	Passed
CpE211	Data Structures and Algorithms	2.00	Passed
CpE221	Numerical Methods	3.00	Passed
CpE222	Software Design	4.00	Passed
CpE311	Logic Circuits and Design	4.00	Passed
CpE312	Operating Systems	3.00	Passed
CpE313	Data and Digital Communications	3.00	Passed
CpE314	Introduction to HDL	1.00	Passed
CpE315	Feedback and Control Systems	3.00	Passed
CpE316	Fundamentals of Mixed Signals and Sensors	3.00	Passed
CpE317	Computer Engineering Drafting and Design	1.00	Passed
CpE321	Basic Occupational Health and Safety	3.00	Passed
CpE322	Computer Networks and Security	4.00	Passed
CpE323	Microprocessors	4.00	Passed
CpE324	Methods of Research	3.00	Passed
CpE325	CpE Laws and Professional Practice	2.00	Passed
CpE413	Emerging Technologies in CpE	3.00	Passed
CpE423	On the Job Training	3.00	Passed
CpEELEC311	Cognate / Elective Course 1**	3.00	Passed
CpEELEC321	Cognate / Elective Course 2**	3.00	Passed
CpEELEC411	Cognate / Elective Course 3**	3.00	Passed
EnE111	Environmental Science and Engineering	3.00	Passed

Republic of the Philippines
SULTAN KUDARAT STATE UNIVERSITY
College of Engineering
Isulan Campus
Isulan, Sultan Kudarat

GE701	Mathematics in the Modern World	3.00	Passed
GE702	Purposive Communication	3.00	Passed
GE703	Ethics	3.00	Passed
GE704	Science, Technology and Society	3.00	Passed
GE705	The Contemporary World	3.00	Passed
GE706	Art Appreciation	3.00	Passed
GE707	Readings in Philippine History	3.00	Passed
GE708	Understanding the Self	3.00	Passed
GE709	The Life and Works of Jose Rizal	3.00	Passed
GE711	Culture of Mindanao	3.00	Passed
GE712	Gender and Society	3.00	Passed
GE713	Kontekstwalisadong Komunikasyon sa Filipino (KOMFIL)	3.00	Passed
GE714	Sosyedad at Literatura / Panitikang Panlipunan (SOSLIT)	3.00	Passed
GE715	Filipino sa Iba't ibang Disiplina (FILDIS)	3.00	Passed
MATH011	Calculus 1	3.00	Passed
MATH012	Calculus 2	3.00	Passed
MATH201	Differential Equations	3.00	Passed
MATH202	Engineering Data Analysis	3.00	Passed
NSTP101	National Service Training Program 1	3.00	Passed
NSTP102	National Service Training Program 2	3.00	Passed
PE101	Physical Fitness and Self-Testing Activities	2.00	Passed
PE102	Rhythmic Activities	2.00	Passed
PE103	Recreational Activities	2.00	Passed
PE104	Team Sports	2.00	Passed
SCI111	Chemistry for Engineers	4.00	Passed
SCI121	Physics for Engineers	4.00	Passed

Total Number of Units Required for the Period : **175 UNITS**
Total Numbers of Units Earned : **159 UNITS**

Certified Correct:

Approved:

JOVANIE S. LINGCAY
Campus Registrar

____
Date

ROMMEL M. LAGUMEN
Campus Director

____
Date

Appendix 3

Republic of the Philippines
SULTAN KUDARAT STATE UNIVERSITY
College of Engineering
Isulan Campus
Isulan, Sultan Kudarat

Form 3

APPLICATION FOR THESIS TITLE

____
Date

Office Automation System for Security through IoT-based Control.

__________ Remarks __________ Signature

Self-Service Printing Station In Sksu – Isulan Library With Cashless Payment Integration Via Gcash

__________ Remarks __________ Signature

LoRa-Enabled Waste Bin Monitoring System for Efficient Waste Management in Schools

__________ Remarks __________ Signature

We are planning to write our thesis outline on August 22, 2019 at SKSU Isulan Campus.

Very respectfully yours,

NATHANIEL C. LAGNAODA

GEORGE ANDRIE D. GATAN
Students

Recommending Approval:

ARNEL N. BEN, MAED
Member

MEILAFLOR A. PACLIBAR, MEP-CoE
Member

JADE A. BUENAVIDES, PhD
Adviser

Endorsed:

ELMER C. BUENAVIDES, DIT.
Campus Research Coordinator

____
Date Signed

LENMAR T. CATAJAY, ME-CoE
College Dean

____
Date Signed

Approved:

ROMMEL M. LAGUMEN
Campus Director

____
Date Signed

Appendix 4

Republic of the Philippines
SULTAN KUDARAT STATE UNIVERSITY
College of Engineering
Isulan Campus
Isulan, Sultan Kudarat

Form 4

NOMINATION OF GUIDANCE COMMITTEE

We, **NATHANIEL C. LAGNAODA** and **GEORGE ANDRIE D. GATAN** are students of **BS in Computer Engineering** hereby nominate the following as adviser and members of my thesis guidance committee.

JADE A. BUENAVIDES, PhD
Adviser

ARNEL N. BEN, MAED
Member

MEILAFLOR A. PACLIBAR, MEP-CoE
Member

We, hereby certify our willingness to act as adviser / members of the guidance committee.

JADE A. BUENAVIDES, PhD
Adviser

ARNEL N. BEN, MAED
Member

MEILAFLOR A. PACLIBAR, MEP-CoE
Member

Endorsed:

ELMER C. BUENAVIDES, DIT
Campus Research Coordinator

____
Date Signed

LENMAR T. CATAJAY, ME-CpE
College Dean

____
Date Signed

Approved:

ROMMEL M. LAGUMEN
Campus Director

____
Date Signed

Appendix 4 Republic of the Philippines Form 4

SULTAN KUDARAT STATE UNIVERSITY

College of Engineering

Isulan Campus

Isulan, Sultan Kudarat

APPLICATION FOR THESIS OUTLINE DEFENSE

Name: **NATHANIEL C. LAGNAODA** Course/Major: **BSCpE**

GEORGE ANDRIE D. GATAN

We have the honor to apply for outline defense for our study entitled: **SELF-SERVICE PRINTING STATION IN SKSU-ISULAN LIBRARY WITH CASHLESS PAYMENT INTEGRATION VIA GCASH**

Time:

Date:

Venue:

ARNEL N. BEN, MAED
Member

MEILAFLOR A. PACLIBAR, MEP-CoE
Member

ARNEL N. BEN, MAED
Statistician

EDRALIN D. MESIAS, PhD
English Critic

JADE A. BUENAVIDES, PhD
Adviser

Endorsed:

ELMER C. BUENAVIDES, DIT
Campus Research Coordinator

Recommending Approval:

LENMAR T. CATAJAY, ME-CoE
College Dean

Approved:

ROMMEL M. LAGUMEN
Campus Director

Appendix 5 Form 5

Republic of the Philippines
SULTAN KUDARAT STATE UNIVERSITY
College of Engineering
Isulan Campus
Isulan, Sultan Kudarat

APPROVAL OF THESIS OUTLINE

Name: **NATHANIEL C. LAGNAODA** Course/Major: **BSCpE**
GEORGE ANDRIE D. GATAN
Major:
Thesis Title: Iot-Based Monitoring System for Calamity Response in the Municipal Agriculture Office in Norala

APPROVED BY THE GUIDANCE COMMITTEE

JADE A. BUENAVIDES, PhD Adviser	____________ Signature	____________ Date
ARNEL N. BEN, MAED Member	____________ Signature	____________ Date
MEILAFLOR A. PACLIBAR MEP-CoE Member	____________ Signature	____________ Date
ARNEL N. BEN, MAED Statistician	____________ Signature	____________ Date
EDRALIN D. MESIAS, PhD English Critic	____________ Signature	____________ Date

Endorsed:

ELMER C. BUENAVIDES, DIT
Campus Research Coordinator

Recommending Approval:

LENMAR T. CATAJAY, ME-CpE
College Dean

Approved:

ROMMEL M. LAGUMEN
Campus Director

Appendix 7 Form 7

Republic of the Philippines
SULTAN KUDARAT STATE UNIVERSITY
College of Engineering
Isulan Campus
Isulan, Sultan Kudarat

CERTIFICATION OF STATISTICIAN

This is to certify that the thesis entitled SELF-SERVICE PRINTING STATION IN SKSU-ISULAN LIBRARY WITH CASHLESS PAYMENT INTEGRATION VIA GCASH conducted on ____________, authored by NATHANIEL C. LAGNAODA and GEORGE ANDRIE D. GATAN was evaluated/checked by the undersigned as to the statistical analysis and interpretation.

Issued on this ______________ day of ______________.

ARNEL N. BEN, MAED
Statistician

Noted:

ROMMEL M. LAGUMEN
Campus Director

Appendix 8

Form 8

Republic of the Philippines
SULTAN KUDARAT STATE UNIVERSITY
College of Engineering
Isulan Campus
Isulan, Sultan Kudarat

CERTIFICATION OF ENGLISH CRITIC

This is to certify that the thesis entitled **SELF-SERVICE PRINTING STATION IN SKSU-ISULAN LIBRARY WITH CASHLESS PAYMENT INTEGRATION VIA GCASH** conducted on ______________ authored by **NATHANIEL C. LAGNAODA** and **GEORGE ANDRIE D. GATAN** was edited by the undersigned as to its grammar.

Issued on this ______ day of _____________.

EDRALIN D. MESIAS, PhD
English Critic

Noted:

ROMMEL M. LAGUMEN
Campus Director

Appendix 9 Form 9

Republic of the Philippines
SULTAN KUDARAT STATE UNIVERSITY
College of Engineering
Isulan Campus
Isulan, Sultan Kudarat

APPLICATION FOR THESIS FINAL DEFENSE EXAMINATION

Name: **NATHANIEL C. LAGNAODA** Course/Major: **BSCpE**
GEORGE ANDRIE D. GATAN
Major: Click here to enter text.
Thesis Title: SELF-SERVICE PRINTING STATION IN SKSU-ISULAN LIBRARY WITH CASHLESS PAYMENT INTEGRATION VIA GCASH

Please write × whether: ( ) First ( ) Second ( )Third
Date: _______________ Time: _________________ Venue: _________

Guidance Committee

Name	Signature	Date
JADE A. BUENAVIDES, Phd Adviser	___________	___________
ARNEL N. BEN, MAED Member	___________	___________
MEILAFLOR A. PACLOBAR, MEP-CoE Member	___________	___________
ARNEL N. BEN, MAED Statistician	___________	___________
EDRALIN D. MESIAS, Phd English Critic	___________	___________

Endorsed: Recommending Approval:

ELMER C. BUENAVIDES, DIT
Campus Research Coordinator

LENMAR T. CATAJAY, ME-CpE
College Dean

Approved:

ROMMEL M. LAGUMEN
Campus Director

____
Date

Republic of the Philippines
SULTAN KUDARAT STATE UNIVERSITY
College of Engineering
Isulan Campus
Isulan, Sultan Kudarat

Report on the Result of Final Defense
(Action taken by the Guidance Committee. Please indicate whether Passed or Failed)

Signature	Date	Remarks
________________	______________________	______________________
________________	______________________	______________________
________________	______________________	______________________

Approved:

ROMMEL M. LAGUMEN
Campus Director

Appendix 10 Form 10

Republic of the Philippines
SULTAN KUDARAT STATE UNIVERSITY
College of Engineering
Isulan Campus
Isulan, Sultan Kudarat

APPLICATION FOR THESIS FINAL PRINTING AND BINDING

This is to certify that the thesis entitled **SELF-SERVICE PRINTING STATION IN SKSU-ISULAN LIBRARY WITH CASHLESS PAYMENT INTEGRATION VIA GCASH** was thoroughly reviewed by the guidance committee and recommended for final printing and binding.

EDRALIN D. MESIAS, PhD
English Critic

____
Date Signed

ARNEL N. BEN, MAED
Statistician

____
Date Signed

ARNEL N. BEN, MAED
Member

____
Date Signed

MEILAFLOR A. PACLIBAR, MEP-CoE
Member

____
Date Signed

JADE A. BUENAIVDES
Adviser

____
Date Signed

Recommending Approval:

ELMER C. BUENAVIDES, DIT
Campus Research Coordinator

____
Date Signed

LENMAR T. CATAJAY, ME-CpE
College Dean

____
Date Signed

Approved:

ROMMEL M. LAGUMEN
Campus Director

____
Date Signed

Appendix 11

SAMPLE QUESTIONNAIRE

RESEARCH INSTRUMENT

SELF-SERVICE PRINTING STATION IN SKSU(SSPS) – ISULAN LIBRARY WITH CASHLESS PAYMENT INTEGRATION VIA GCASH

Name (Optional): ____

Instruction: Rate the self-service printing station (SSPS) from 1 to 5 according to rating scale. Check the table which correspond to your evaluation.

Rating scale to be used in evaluating the SSPS system

Rating	**Interpretation**
5	Strongly Agree
4	Agree
3	Moderately Agree
2	Disagree
1	Strongly Disagree

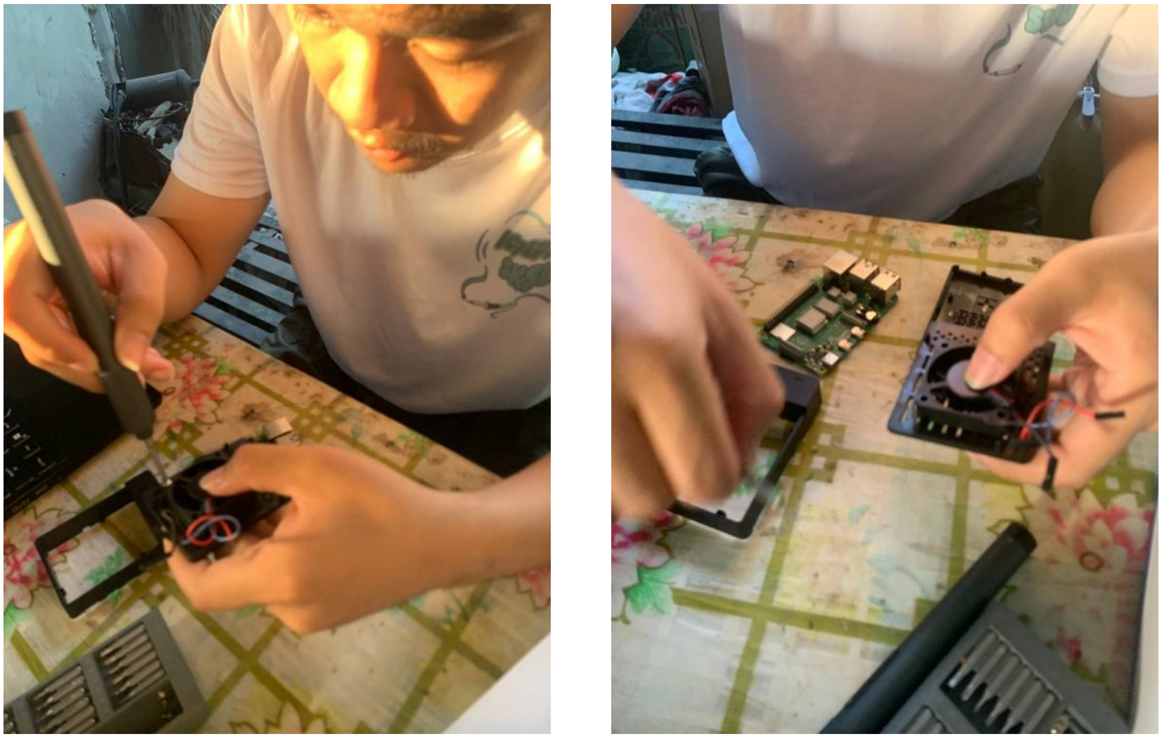
IV. Functionality					
Questions	**Ratings**				
The SSPS system is functional in terms of:	**5**	**4**	**3**	**2**	**1**
6. The SSPS System boots without error.					
7. The system interface is responsive and user-friendly					
8. The file uploading operates smoothly.					
9. The GCash payment process functions properly.					
10. The system completes printing accurately.					

V. Reliability					
Questions	**Ratings**				
The SSPS system is reliability in terms of:	**5**	**4**	**3**	**2**	**1**
6. The system consistently performs printing and payment transactions without failure.					
7. The SSPS responds promptly to user actions without delays or unresponsiveness					
8. The SSPS remains functional and accessible throughout library hours.					
9. Payments made through GCash are accurately verified and recorded.					
10. The SSPS provides accurate output according to the user's selected file.					

VI. User Convenience					
Questions	**Ratings**				
The SSPS system is user convenience in terms of:	**5**	**4**	**3**	**2**	**1**
6. The SSPS is easy to navigate and understand even for first-time users.					
7. The cashless payment option through GCash makes transactions faster and easier.					
8. The SSPS is accessible and available for use when needed.					
9. The SSPS allows me to complete printing tasks quickly and efficiently.					
10. Overall, the SSPS provides a convenient and satisfying user experience.					

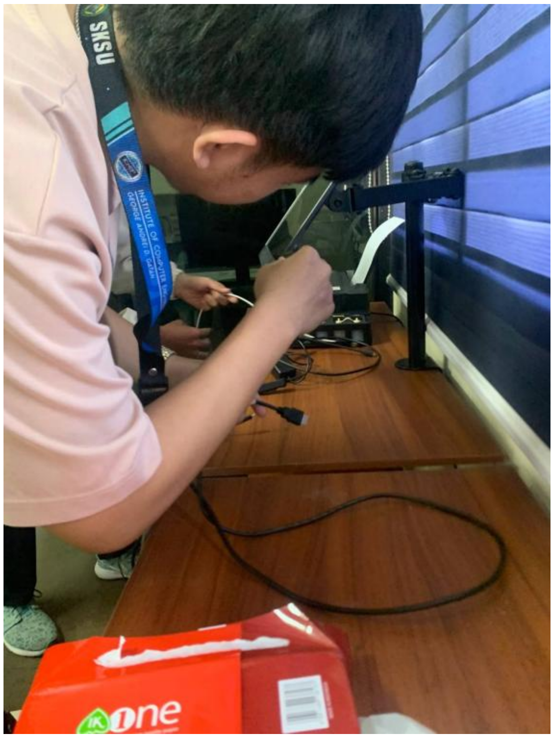
Appendix 12

PICTORIALS AND SCREENSH



Actual Photo of Assembling Raspberry Pi with Enclosure

Actual Photo of Developing the System Code



Actual Photo of Installing The SSPS

``

Actual Photo of Orientation of The System (Library Staff)

Actual Photo of Orientation of The System (Students)

HF	Q1	Q2	Q3	Q4	Q5	HR	Q1	Q2	Q3	Q4	Q5	HC	Q1	Q2	Q3	Q4	Q5
1	5	5	5	5	5		5	4	5	5	5		5	5	5	5	5
2	5	5	5	5	5		5	5	5	5	5		5	5	5	5	5
3	5	5	5	5	5		5	5	5	5	5		5	5	5	5	5
4	4	4	4	4	4		4	4	4	4	4		4	4	4	4	4
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Mean	4.58	4.65	4.605	4.645	4.625	4.621	4.605	4.635	4.625	4.645	4.64	4.63	4.615	4.655	4.635	4.66	4.675
Standard Deviatio	0.513	0.493	0.503	0.489	0.494		0.503	0.492	0.494	0.489	0.491		0.498	0.486	0.492	0.484	0.476

SCREEN SHOT OF THE EVALUATION COMPUTATION

Appendix 13

SOURCE CODE

Appendix 14

DVD